

EMLyon Business School
Professional Project — Group 6

Empirical Asset Pricing via Machine Learning

A Replication and Extension of Gu, Kelly & Xiu (2020)

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1. Extended Introduction

1.1 Background and Relevance of the Topic

Understanding and predicting stock returns has long been one of the central challenges in empirical asset pricing. For several decades, financial economists have attempted to identify variables capable of explaining the cross-section of stock performance and forecasting future returns. Traditional approaches have largely relied on linear factor-based frameworks, ranging from the Capital Asset Pricing Model to multifactor extensions such as the Fama-French models. While these approaches have generated major insights into the determinants of expected returns, they have also highlighted the intrinsic difficulty of the problem: financial return predictability is typically weak, noisy, and highly sensitive to specification choices.

These limitations become even more visible in high-dimensional environments. Financial markets generate a large amount of information, and expected returns may depend simultaneously on firm-specific characteristics, market conditions, and macro-financial variables. In such settings, traditional econometric models often struggle to fully exploit the richness of the available information. Linear models in particular impose restrictive assumptions and may fail to capture nonlinear relationships, threshold effects, and interaction structures that can be economically meaningful.

The rise of machine learning has therefore renewed interest in empirical asset pricing. Machine learning methods are specifically designed to process large predictor sets, handle complex interactions, and detect nonlinear patterns in the data. These properties make them particularly attractive in finance, where the predictive signal is weak and often embedded in a complex cross-sectional structure. The project brief itself is explicitly framed in this perspective, describing the study as an exercise in empirical asset pricing via machine learning and linking prediction directly to the construction of profitable long/short strategies.

This research question is not only academically relevant but also highly important from a practical asset-management perspective. In professional portfolio management, the value of a model does not primarily come from predicting exact return levels; it comes from ranking securities in a way that improves investment decisions. A model may therefore remain useful even when conventional forecasting metrics appear modest, provided that it helps distinguish likely outperformers from likely underperformers. This is why the economic evaluation of predictive models through portfolio construction is central to the logic of this report.

One of the most influential references in this field is *Empirical Asset Pricing via Machine Learning* by Gu, Kelly, and Xiu (2020), which demonstrates that machine learning models can improve stock return prediction relative to traditional linear methods and can generate economically meaningful trading strategies when applied to large datasets of firm characteristics and macroeconomic predictors. The present project is directly inspired by this intellectual framework and seeks to assess whether similar insights can be reproduced and interpreted in a structured empirical setting.

1.2 Research Problem

Despite decades of research in empirical asset pricing, predicting stock returns remains an extremely challenging task. Financial markets are characterized by a high level of noise, rapidly changing economic conditions, and complex interactions between firm-specific characteristics and macroeconomic factors. As a result, identifying reliable predictors of future returns has proven to be difficult, and the statistical signal contained in financial data is often relatively weak.

Traditional empirical asset pricing models generally rely on linear regression frameworks that relate expected returns to a set of observable characteristics. While these approaches have been instrumental in identifying several well-known anomalies, such as size, value, or momentum effects, they may be insufficient when dealing with large sets of predictors and complex nonlinear relationships. In particular, linear models impose strong structural assumptions that may limit their ability to fully capture the underlying dynamics of financial markets.

In recent years, machine learning techniques have emerged as a promising alternative to traditional econometric models. These methods offer greater flexibility and are capable of handling high-dimensional datasets, automatically selecting relevant variables, and capturing nonlinear patterns in the data. As a result, they may provide a more powerful framework for predicting stock returns and identifying investment opportunities. However, an important question remains: do machine learning models truly improve the prediction of future stock returns in a way that is economically meaningful? In other words, even if machine learning methods achieve better statistical fit, it is essential to determine whether these improvements translate into profitable investment strategies.

This project addresses this question by investigating the ability of several predictive models to forecast stock returns at two different horizons: one month and twelve months ahead. The analysis compares traditional linear methods with more flexible nonlinear algorithms, including tree-based models and neural networks. By evaluating both the statistical performance of these models and the economic value of the investment strategies derived from their predictions, the project aims to assess whether machine learning techniques provide a significant advantage over conventional approaches in empirical asset pricing. In particular, the predictive models are used to rank stocks according to their expected returns. These rankings are then translated into long-short portfolios that buy stocks with the highest predicted returns and sell those with the lowest predicted returns. The performance of these portfolios provides a direct measure of the economic usefulness of the predictive models. Through this framework, the project seeks to determine whether machine learning models can effectively extract predictive signals from financial data and whether these signals can be transformed into profitable trading strategies in an out-of-sample setting.

1.3 Objectives, Empirical Strategy, and Key Contributions

This report is structured around a sequential empirical framework designed to bridge the gap between pure statistical forecasting and real-world economic utility in asset pricing. The primary objective is to evaluate the comparative advantages of various machine learning architectures, ranging from regularized linear models (Lasso) to complex non-linear learners such as decision trees (Random Forest, GBRT) and deep neural networks, for extracting predictive signals in a high-dimensional data environment.

At the heart of our approach lies a chronological estimation strategy that progresses from a baseline fixed-split parameterization toward a rigorous 3-way split train-validation-test procedure. This protocol allows us not only to identify the best-performing models but also to quantify how sensitive their predictive effectiveness is to algorithmic calibration and hyperparameter tuning. By subsequently re-applying the optimal parameter combinations to the initial fixed-split framework, we assess the stability and generalizability of these machine learning configurations across a test period that includes critical market phases.

A distinctive contribution of this study is the integrated treatment of macroeconomic conditioning. Unlike approaches that treat macroeconomic variables as secondary extensions, this report incorporates macro-financial interactions, specifically with Dividend-Price and Earnings-Price ratios, directly into the feature space from the very beginning of the empirical study. This design enables a deeper analysis of how different algorithms exploit an enriched predictive environment to modulate risk premiums according to the broader economic context.

Finally, we explicitly distinguish between statistical accuracy (out-of-sample) and economic value (Sharpe Ratio). We demonstrate that models with modest or even negative statistical fit can still generate superior investment strategies through effective cross-sectional security ranking. Our findings highlight the outperformance of the NN3 architecture, which achieves a Sharpe Ratio of 1.21 at the 1-month horizon, as well as the efficiency of the GBRT model, whose Sharpe ratio improves R^2 from 0.68 to 0.92 following optimization. This dual-horizon analysis (1-month and 12-month) ensures a comprehensive evaluation of signal persistence across different investment cycles.

1.4 Structure of the Report

The report is organized as a progressive empirical investigation of stock return predictability and its economic value in portfolio construction. It begins by establishing the methodological framework of the study, including the structure of the feature space, the model families under consideration, the portfolio formation logic, and the evaluation criteria used to assess both statistical and economic performance.

It then turns to the baseline fixed-split analysis, which provides the first benchmark comparison across models and forecast horizons. This initial stage is followed by a validation-based tuning procedure, designed to examine how far model performance depends on hyperparameter selection and on the structure of the estimation framework itself. Once the preferred parameter combinations have been identified, the report returns to the original fixed-split design in order to test whether these optimized specifications actually improve the baseline results when reintroduced into the initial empirical setting.

The empirical analysis is then extended to neural networks, which represent an additional layer of modeling flexibility and make it possible to test whether deep architectures can uncover predictive signals that remain less accessible to linear and tree-based methods. At this stage, the report deliberately postpones any final cross-model ranking in order to preserve a clear distinction between model-specific analysis and broader comparative synthesis.

After these sequential empirical sections, the report shifts from model-by-model analysis to a broader comparative perspective. It synthesizes the results across model families, estimation settings, horizons, and portfolio strategy types, with particular attention to the distinction between long, bottom-decile, and long-short implementations, as well as to the recurrent tension between statistical fit and economic value.

The final part of the report develops a broader discussion of the main lessons of the study. It examines the implications of the ranking-based nature of predictive success, the role of macro-conditioned interactions, the asymmetry between long and short portfolio construction, and the operational limitations associated with more flexible machine learning models. The report concludes by restating the central findings, discussing their implications for empirical asset pricing and systematic portfolio management, and opening the discussion toward future research directions and more advanced quantitative applications.

2. Methodological Framework and Evaluation Design

2.1. Research Design and Empirical Timeline

The core objective of this study is to evaluate the effectiveness of machine learning (ML) architectures in identifying cross-sectional stock return patterns and successfully monetizing them through profitable investment strategies. The research design is structured to move beyond a purely statistical forecasting exercise by explicitly testing the economic value of these predictions in an out-of-sample investment context.

The empirical investigation is conducted across two distinct forecast horizons: 1-month forward returns and 12-month forward returns. This dual-horizon framework is methodologically critical, as it allows for a comparative analysis of short-term noise versus medium-term structural signals. While 1-month predictability is often characterized by a lower signal-to-noise ratio, the 12-month horizon provides insight into the persistence of risk premiums and the stability of characteristic-based signals.

To ensure a disciplined and bias-free evaluation, the study follows a rigorous chronological workflow divided into four distinct stages:

1. **Baseline Estimation (Fixed-Split):** Establishing initial benchmarks for OLS, Lasso, Random Forest, and GBRT using light, robust hyperparameter settings and a simple train-test partition.
2. **Hyperparameter Tuning (3-Way Split):** Implementing a training-validation-test structure to optimize model configurations and identify the most effective architectures for each horizon.
3. **Optimized Fixed-Split Re-estimation:** Applying the winning hyperparameter combinations back to the original fixed-split framework to isolate the specific gains generated by the optimization process.
4. **Neural Network Extension:** Expanding the analysis to deep learning architectures (NN3) to capture high-order non-linearities that may remain inaccessible to linear and tree-based models.

The dataset (1960-2021) is partitioned into three disjoint periods to strictly prevent look-ahead bias and ensure the validity of the out-of-sample results:

- **Training Period (1960-1990):** Used for initial model estimation and the learning of parameter weights.
- **Validation Period (1990-2005):** Dedicated to the systematic tuning of hyperparameters and model selection based on intermediate performance.
- **Testing Period (2005-2021):** Reserved for the final performance evaluation. This period is particularly demanding as it encompasses diverse market regimes, including the 2008 financial crisis and the 2020 pandemic-induced volatility.

2.2. Model Families and Feature Space

2.2.1. A High-Dimensional Predictor Space

The structure of the information used in this study relies on a massive dataset comprising 94 company-specific features (e.g., momentum, size, volatility, liquidity). Unlike traditional econometric approaches, which are often limited to a small number of pre-established factors, our methodology embraces high dimensionality to allow algorithms to detect subtle signals in the cross-section of assets.

2.2.2. Integrated Macro-Financial Conditioning

The fundamental characteristic of our empirical setup is that macroeconomic information is incorporated into the feature space from the very beginning of the study. Rather than introducing macroeconomic variables as isolated predictors in a later phase, we generated interaction terms between firm-level characteristics and macro-financial indicators (such as Dividend-Price and Earnings-Price ratios) starting from the baseline estimations.

This design choice is methodologically significant: it allows the models to exploit not only company characteristics in isolation, but also the dependence of these characteristics on the global macroeconomic environment. Thus, any divergence in performance between the models cannot be attributed to the presence or absence of macroeconomic information, but rather to the capacity of each algorithm to process and interpret this enriched environment.

2.2.3. Algorithmic Diversity and Rationale for Families

The comparison protocol relies on four families of models representative of the state of the art in machine learning applied to finance:

- Linear Benchmarks (OLS and Lasso): OLS provides the simplest and most transparent point of comparison. The Lasso (Least Absolute Shrinkage and Selection Operator) extends this framework by adding an L_1 regularization, allowing for both variance reduction and the automatic selection of the most relevant variables.
- Non-Linear Ensemble Methods (Random Forest and GBRT): These models are introduced to capture non-linear relationships, threshold effects, and complex interaction structures that linear additive models cannot approximate.
 - Random Forest relies on the aggregation (bagging) of multiple decision trees to stabilize predictions.
 - GBRT (Gradient Boosted Regression Trees) builds trees sequentially, with each new iteration aiming to correct previous prediction errors.

This diversity allows for testing the central hypothesis that the predictability of stock returns is better captured by flexible functional forms capable of navigating a vast and noisy predictor space.

2.3. Data Processing Strategy and Learning Optimization

2.3.1. The Data Volume Challenge

The processing of a panel covering the 1960-2021 period, enriched by more than 90 fundamental characteristics and their macroeconomic interactions, generates a massive volume of data that exceeds conventional processing capacities for non-linear models. In order to maintain statistical rigor while respecting realistic computational constraints, we have implemented a differentiated sampling and learning strategy based on the mathematical nature of the algorithms.

2.3.2. Stratified Annual Sampling Method: The «Distributed 10% Rule»

Rather than using a compact but limited time window, we prioritized a stratified annual sampling approach. For each year in the sample, a random fraction of the data is selected. This method offers a major theoretical advantage: it ensures that the models are exposed to an exhaustive diversity of economic cycles, from the 1973 oil shocks to the 2008 financial crisis, including technological bubbles and varied inflation regimes.

This “trans-temporal” exposure allows the algorithms to distinguish persistent predictive signals from transitory noise, granting the models a statistical resilience far superior to that obtained through exhaustive learning over a short and specific period.

2.3.3. Technical Configuration and Learning Rates

The sampling rates were calibrated specifically for each model family to maximize the density of the useful signal:

- Linear Models (OLS/Lasso): These models are trained on 100% of the data. By using Stochastic Gradient Descent (SGD) and the «partial_fit» method, they process the data stream iteratively, allowing for exhaustive learning without saturating the RAM.
- Ensemble Methods (Random Forest and GBRT): During the Fixed Split (baseline) phase, a 10% annual sampling rate is used to balance signal richness and execution speed.
 - During the Hyperparameter Optimization (Tuning) phase, this rate is reduced to 5% to allow for the rapid exploration of hundreds of parameter combinations.
- Neural Networks (Deep Learning): The Tuning phases use 10% for training and 20% for validation.
 - The final optimized model uses 20% of the data. This increased volume is essential to allow the neurons to specialize and capture the complex non-linear interactions between variables, thereby maximizing the final predictive power.

This hierarchy ensures that each model receives the amount of information suited to its complexity: «brute force» for linear models and signal density for Deep Learning.

2.4. Optimization Methodology and Search Space (Hyperparameter Tuning)

2.4.1. Objective of the Validation Phase

The optimization phase (or Tuning) constitutes a pivotal stage of our experimental protocol. Its objective is to move beyond the initial «baseline» parameters deliberately chosen for their lightness to identify the algorithmic configurations that maximize predictive power on data unseen during training. This search is conducted exclusively over the Validation period (1990-2005), thereby guaranteeing the integrity of the final out-of-sample evaluation.

2.4.2. Search Space Configuration

For each model family, we defined a search space (parameter grid) targeting the most critical performance and regularization levers:

Linear Regularization (Lasso): The focus is on the strength of the L_1 penalty alpha. We tested a logarithmic scale (ranging from 10^{-5} to 10^{-1}) to identify the optimal level of parsimony, capable of eliminating noise while retaining the most persistent predictors.

Ensemble Models (RF and GBRT): The search focused on the bias-variance tradeoff via two parameters:

- **Maximum depth (max_depth):** Explored from 10^{-1} (often sufficient for a 12-month horizon) up to 10^{-1} to capture finer interactions at 1 month.
- **Number of estimators (n_estimators):** Tested up to 300 to stabilize the Random Forest predictions.
- **Neural Networks (Deep Learning):** The search space focused on architecture and learning dynamics:
- **Complexity:** Comparison of NN1 (1 layer), NN2 (2 layers), and NN3 (3 layers) structures.

Activation function: Comparison of the historical Tanh (Hyperbolic Tangent) activation against the ReLU (Rectified Linear Unit) function, the latter being preferred for its ability to accelerate convergence and limit the risk of vanishing gradients.

2.4.3. Computational Constraints and Agility

Exploring this search space, involving hundreds of combinations, requires rigorous management of computing resources. As detailed previously, we reduced the sampling rate to 5% per year for tree-based models and to 10% for neural networks during this specific phase. This agility allows for an exhaustive exploration of the parameter grid without sacrificing the temporal representativeness necessary to identify robust configurations.

2.4.4. Traceability of Optimal Parameters

At the conclusion of this phase, the combinations offering the best performance metrics are systematically saved in a registry (best_parameters.csv). These parameters are then re-injected into the final model for evaluation over the test period (2005-2021), thus ensuring total traceability between the research phase and the results presented in this report.

2.5. Portfolio Formation Logic and Evaluation Metrics

2.5.1. Ranking Mechanism and Portfolio Construction

The transition from statistical prediction to investment strategy relies on a cross-sectional ranking mechanism. For each model and at each time step t , the predicted returns $\widehat{r_{i,t+1}}$ for all securities are sorted in descending order. Based on this ranking, the assets are divided into deciles:

- Long Portfolio: Composed of securities belonging to the top decile (the top 10% of the highest predictions).
- Short Portfolio: Composed of securities belonging to the bottom decile (the 10% of the lowest predictions).
- Long-Short (L-S) Strategy: Constructed by taking a long position on the Long portfolio and a short position on the Short portfolio. This approach allows for isolating the alpha generated by the model while partially neutralizing exposure to global market movements.

2.5.2. Primacy of Ranking Over Point Accuracy

A conceptual pillar of this study is that the economic value of a model in finance does not depend on its ability to predict the exact level of a return, but on its ability to correctly prioritize assets. A model may exhibit a low, or even negative, out-of-sample while remaining extremely profitable if it manages to maintain a systematic and positive gap between the performances of the extreme deciles. The Long-Short portfolio is therefore the most informative object of analysis, as it crystallizes the algorithm's ability to distinguish future winners from future losers.

2.5.3. Statistical Performance Indicators

Statistical evaluation relies primarily on the out-of-sample (R^2_{OOS}). It measures the reduction in mean squared error relative to a benchmark (generally the historical mean of returns). Given the intrinsic volatility of financial markets, an R^2_{OOS} even very close to zero (or slightly positive) is considered evidence of significant predictability in contemporary academic literature.

2.5.4. Economic Performance Metrics

To evaluate the financial viability of the strategies, we use three key risk-adjusted return indicators:

- Annualized Return: Measures the gross profitability of the strategy on an annual basis.
- Annualized Volatility: Quantifies the risk or uncertainty associated with the strategy's returns.
- Sharpe Ratio: The final arbitrage criterion, calculated as the ratio between excess return and volatility. It allows for comparing the efficiency of models (linear vs. non-linear) on a normalized basis.

2.5.5. Cumulative Wealth Analysis

Finally, the study relies on cumulative wealth curves (equity curves). These graphical representations are essential for diagnosing the stability of performance over time. They allow for identifying whether a model's gains are constant or if they result from a few specific episodes, and for evaluating the strategy's resilience in the face of systemic crises (e.g., 2008, 2020).

2.6. Reproducibility and Technical Determinism

2.6.1. The Pillar of Quantitative Auditing

In the fields of quantitative finance and machine learning, reproducibility is not merely a technical requirement but a fundamental pillar of scientific integrity. Given the stochastic nature of many optimization algorithms, ensuring that performance metrics and diagnostics can be fully audited and replicated is essential. To this end, a strict deterministic protocol was enforced throughout the entire empirical pipeline.

2.6.2. Systematic Randomness Control: The «Seed 42» Protocol

The training of modern predictive models particularly Random Forests and Deep Neural Networks rely on stochastic processes for weight initialization, data shuffling, and feature selection. To neutralize variations caused by randomness, a universal Random Seed (42) was systematically applied across the software infrastructure:

- Numerical Operations: `np.random.seed(42)` ensures consistent array manipulations and data transformations.
- Algorithmic Frameworks: The `random_state=42` parameter was strictly defined in all Scikit-Learn modules, including `SGDRegressor`, `RandomForestRegressor`, and `GradientBoostingRegressor`.
- Stochastic Sampling: The stratified sampling method utilized `sample(frac=X, random_state=42)`, guaranteeing that every model «sees» the exact same data points across different executions.

2.6.3. Hardware and Software Determinism

Beyond software seeds, additional constraints were imposed on high-performance computing components to mitigate micro-variations in floating-point calculations:

- **CUDA (GPU) Determinism:** For Neural Networks trained on NVIDIA hardware via PyTorch, the flags `torch.backends.cudnn.deterministic = True` and `torch.backends.cudnn.benchmark = False` were activated. This prevents the GPU from selecting non-deterministic optimization algorithms that could lead to slight variations in final weights.
- **Standardized Environment:** All analyses were conducted within a sealed folder structure (`Réponses_aux_questions`) using a reconstructed Parquet dataset. This ensures that the data reading path and pre-processing steps remain invariable.

2.6.4. Traceability and Parameter Documentation

To ensure total transparency, every result presented in this report is linked to a documented technical artifact:

- **Best Parameters Log:** The optimal hyperparameters identified during the validation phase (Sections 3 and 4) are dynamically saved in `best_parameters.csv`. This file serves as the unique source of truth for final model re-estimations.
- **Diagnostic Reports:** Every execution generates a dedicated `.txt` log containing detailed metrics (Sharpe Ratio, R^2 , Annualized Returns). These logs provide an immutable audit trail, allowing for an instant verification of the theoretical results against the actual code execution journals.

3. Baseline Fixed-Split Estimation and Initial Findings

3.1. Baseline Setup and Hyperparameter Logic

The initial empirical stage establishes a «Baseline» benchmark to evaluate the raw predictive power of each model family before systematic optimization. We employ a fixed-split protocol, where models are trained on the 1960-2004 period and evaluated on the 2005-2021 out-of-sample (OOS) window.

In line with the approach of Gu, Kelly, and Xiu (2020), we begin with lightweight and robust hyperparameter configurations. This choice is methodologically deliberate: it allows us to isolate the «default» behavior of each algorithm in a high-dimensional space conditioned by macro-interactions. While Gu et al. utilize rolling-window validation to update parameters, our baseline serves as a static stress test to identify which architecture inherently captures the cross-sectional signal without extensive calibration.

Table 1: Summary of Baseline Hyperparameters

Model	Hyperparameter	Value
OLS	Penalty	0
	Optimizer	SGD
	Initial Learning Rate	0.01
Lasso	Penalty	L1
	Alpha	10^{-5}
	Initial Learning Rate	0.01
Random Forest	Number of trees	100
	Max Depth	3
	Seed (Random State)	42
GBRT	Number of iterations	50
	Max Depth	2
	Learning Rate	0.1
	Seed (Random State)	42

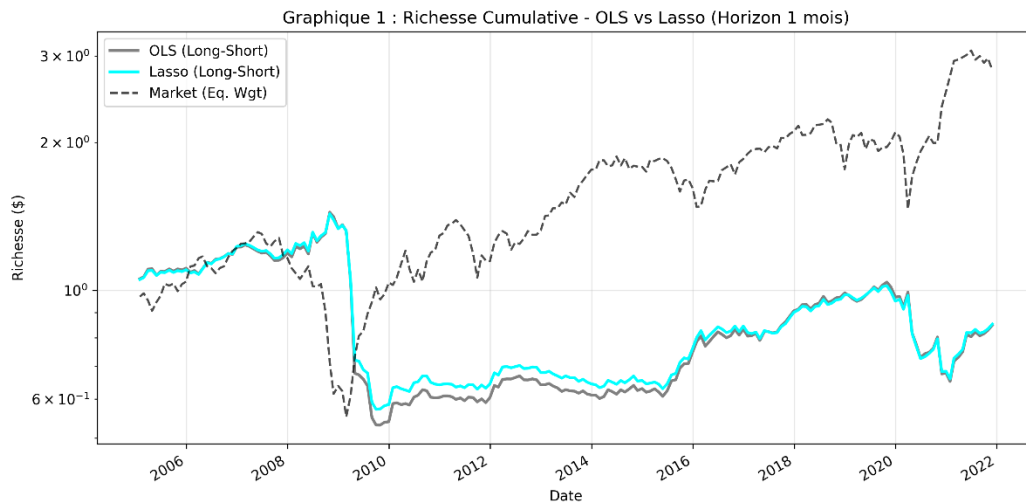
3.2. The 1-Month Horizon: Ranking Power vs. Statistical Noise

The one-month forecast horizon (ret) constitutes the most demanding environment of this study due to an extremely low signal-to-noise ratio. Our baseline results reveal a fundamental divergence between statistical accuracy metrics R_{OOS}^2 and actual economic utility (Sharpe Ratio).

3.2.1. The Failure of Linear Models (OLS and Lasso)

A striking result of the baseline exercise is the near-perfect identity between the performances of OLS and Lasso. Both models display strongly negative R_{OOS}^2 (below -5.8%) and Sharpe ratios close to zero (~0.02).

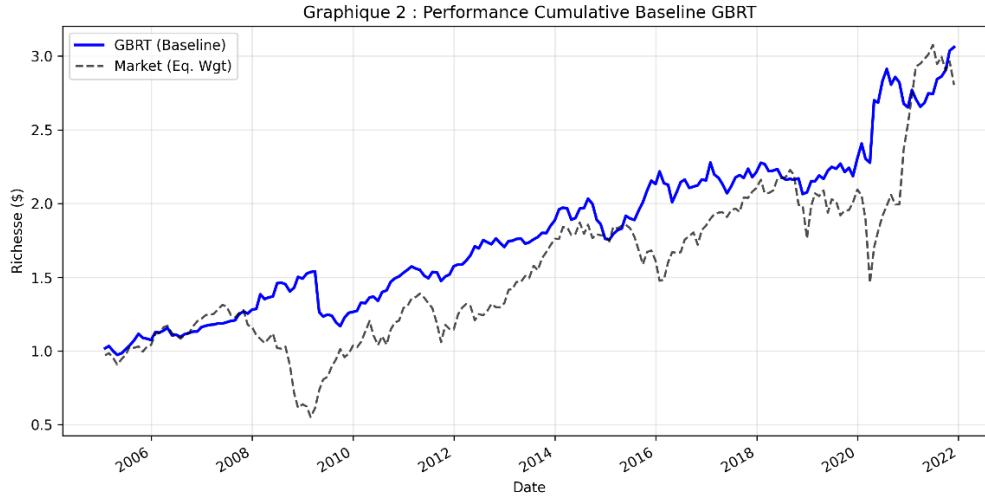
This suggests that at a one-month horizon, the major challenge is not overfitting which the Lasso's L_1 penalty is supposed to mitigate but rather a lack of expressive capacity. Linear models, even when enriched with macroeconomic interactions, fail to capture the nonlinearities and threshold effects necessary to separate future winners from future losers. Contrary to the study by Gu, Kelly, and Xiu (2020), where Lasso proves competitive once optimized, our baseline confirms that without fine-tuning parsimony, the linear framework remains «blind» to the complexity of the 94-feature space.



3.2.2. GBRT: Economic Dominance through Sequential Learning

GBRT emerges as the superior benchmark model, achieving a Sharpe ratio of 0.6845. Although its R^2_{OOS} remains negative (-1.03%), its cumulative wealth curve (Graph 2) displays a persistent upward trend.

This result corroborates the central conclusion of the reference study: Boosting is exceptionally effective at ranking assets. By building trees sequentially to correct previous errors, GBRT focuses on the «marginal» signal, which is crucial in finance. The model succeeds in clearly separating the extreme deciles, delivering an annualized Long-Short return of 7.18% with a volatility of 10.49%.



3.2.3. The Random Forest Paradox: A Statistical Mirage

The Random Forest (RF) results present the richest methodological lesson. Statistically, RF appears to be the «best» model with an R^2_{OOS} of -0.1035% (the closest to zero). Yet it represents a total economic failure, with a negative Sharpe ratio of -0.3545.

Graphical analysis reveals a misalignment in ranking: the bottom decile portfolio (supposed to group the losers) systematically outperformed the top decile (the winners). This «mirage» occurs because the bagging process (averaging deep trees) can reduce statistical variance but lose the directionality of the signal if the base trees are not calibrated. While Gu et al. (2020) place RF among the top models, our baseline shows that without optimization of depth d and sampling, RF can be more misleading than simpler models.

Table 2: Comparative Synthesis of 1-Month Baseline Performance

Model	R^2_{OOS}	L-S Return	L-S Volatility	L-S Sharpe
GBRT	-1.03%	7.18%	10.49%	0.6845
Lasso	-5.88%	0.16%	14.11%	0.0114
OLS	-7.08%	0.33%	15.18%	0.0217
RF	-0.10%	-3.22%	9.09%	-0.3545

3.3. The 12-Month Horizon: From Noise to Structural Signal

Extending the forecast horizon to 12 months (ret12) profoundly modifies the dynamics of the results. While the monthly horizon is dominated by noise and short-term reversals, the annual horizon allows fundamental signals (valuation, profitability) and macroeconomic interactions to gain persistence. For the first time in this baseline study, we observe the emergence of positive R^2_{OOS} , confirming that the structure of medium-term returns is more statistically predictable.

3.3.1. The Statistical Turning Point of Random Forest (RF)

The most notable result at this horizon is the rehabilitation of Random Forest. While it was economically failing at one month, RF here becomes the «statistical champion» with a positive R^2_{OOS} of 2.31%.

This spectacular improvement suggests that the bagging logic (averaging of trees) is particularly suited to slow and persistent signals. By averaging predictions over a long horizon, RF succeeds in filtering out idiosyncratic noise to capture structural trends. On an economic level, the model finally aligns it delivers a Long-Short return of 3.52% and a Sharpe ratio of 0.2971. Although respectable, this statistical success is not yet enough to dethrone GBRT in the field of risk-adjusted efficiency.

3.3.2. GBRT: Remarkable Consistency in Ranking

GBRT confirms its robustness by remaining the economic leader with the highest Sharpe ratio (0.4254). Although its R^2_{OOS} is slightly negative (-0.16%), its ability to generate a Long-Short spread of 5.51% (the highest in the baseline category) demonstrates that sequential learning (boosting) remains the most powerful prioritization tool, regardless of the horizon.

As emphasized by Gu, Kelly, and Xiu (2020), tree-based models dominate linear models because they better exploit complex non-linear interactions between firm characteristics and the macroeconomic cycle. At 12 months, this superiority is crystallized by better-controlled Long-Short volatility (12.94%) compared to linear models.

Table 3: Comparative Synthesis of 12-Month Baseline Performance

Model	R^2_{OOS}	L-S Return	L-S Volatility	L-S Sharpe
RF	2.31%	3.52%	11.86%	0.2971
GBRT	-0.16%	5.51%	12.94%	0.4254
OLS	-8.78%	5.15%	17.40%	0.2961
Lasso	-8.51%	5.11%	17.37%	0.2941

3.3.3. The Relative Resilience of Linear Models (OLS and Lasso)

At the 12-month horizon, linear models (OLS and Lasso) see their economic performance improve significantly, reaching Sharpe ratios of approximately 0.30. They produce Long-Short returns exceeding 5%, approaching GBRT in terms of gross profitability.

However, their statistical accuracy remains mediocre with strongly negative R^2_{OOS} (around -8.5%). This contrast indicates that while OLS and Lasso manage to capture the general direction of one-year return trends, they fail completely to predict the magnitude of movements (level prediction). Once again, the near-perfect identity between OLS and Lasso shows that L_1 regularization brings no substantial benefit within a linear framework incapable of modelling the complex interactions present in the data.

3.3.4. Synthesis of the 12-Month Baseline Hierarchy

The model hierarchy at 12 months is more balanced than at one month, but the fundamental conclusions remain:

- RF is the most statistically accurate, validating the existence of an exploitable structural signal at one year.
- GBRT remains the most economically efficient, proving that boosting is the best algorithm for Long-Short portfolio construction.
- Linear models lack flexibility, remaining unable to compete with non-linear architectures for capturing the conditional risk premium.

3.4. Variable Importance: What is the Baseline Learning?

Beyond the numerical performance, the primary interest of this study lies in identifying the economic sources of predictability. Feature Importance analysis allows us to understand how each model family prioritizes information within our high-dimensional feature space. Our results highlight two distinct learning regimes: a linear regime focused on local price signals and a non-linear regime exploiting the macro-conditioned structure.

3.4.1. The Prism of Linear Models: A Dependency on Momentum

The OLS and Lasso models display a massive concentration on Momentum signals and their macroeconomic interactions. Variables such as `mom1m`, `mom6m`, and `mom12m`, often multiplied by global valuation ratios (`ep` or `dp`), systematically dominate the ranking.

This dependency on momentum partly explains the weakness of linear models at the one-month horizon: by focusing on past price trends, these models struggle to capture abrupt reversals or more subtle interactions related to the fundamental health of companies. As observed by Gu, Kelly, and Xiu (2020), classical regressions tend to «overreact» to the most visible short-term signals, neglecting slower valuation variables that require a more flexible modelling structure.

3.4.2. GBRT and RF: Exploiting Macro-Conditioned Interactions

Conversely, tree-based models (GBRT and RF) demonstrate a superior ability to extract value from fundamental interactions. The most powerful signal identified by GBRT at the one-month horizon is the interaction between the dividend payout ratio and the Treasury bill rate (`divo_x_macro_tbl`), representing more than 16% of the total feature importance.

This predominance of interactions suggests that non-linear models do not merely add predictors but truly learn to modulate the importance of a financial characteristic based on the interest rate regime or the economic cycle. Other variables such as «sin» stocks (`sin`) and valuation measures (`divi`, `bm`) also appear as pillars of the ranking, confirming that boosting and bagging rely on a much broader and more diverse information base than linear models.

3.4.3. Evolution of the Hierarchy by Horizon (1m vs. 12m)

A key takeaway from this section is the shift in feature importance as the horizon lengthens. At 1 month, the signal is a complex mixture of macroeconomic interactions and liquidity. At 12 months, we observe a «purification» of the signal: pure fundamental variables, such as divo (24.9% importance for GBRT at one year) and the sin stock signal (13.3%), take the lead.

This finding validates our analysis in Section 3.3: as short-term noise fades, non-linear models fall back on structural value indicators, thus explaining the shift to positive R_{OOS}^2 and better stability in Long-Short portfolios.

3.4.4. Comparison with Reference Literature

Our baseline results partially confirm the conclusions of Gu et al. (2020), who noted that trees and neural networks grant increased importance to liquidity and valuation characteristics. However, our study highlights an even more central role for direct macroeconomic interactions, proving that enriching the feature space from the baseline phase is a major performance driver for non-linear algorithms.

3.5. Synthesis and Transition to Optimization

The baseline exercise allows us to draw three fundamental conclusions for the remainder of the study.

First, the choice of model is decisive from the earliest stages of the analysis: algorithms are not interchangeable and produce distinct performance hierarchies. Second, GBRT emerges as the most economically robust model, delivering the best risk-adjusted Sharpe ratios across both time horizons. Third, the near identity of the results between OLS and Lasso suggests that the weakness of linear models does not stem from a lack of regularization, but from a structural inability to capture the non-linear predictive relationships and complex interactions present in financial data.

A broader methodological lesson also emerges: the disconnection between statistical fit and financial utility. The success of GBRT demonstrates that a model can generate superior Long-Short returns despite a negative R_{OOS}^2 , while the case of Random Forest at one month illustrates the opposite risk: an acceptable predictive score can mask a poor economic alignment of asset rankings. This distinction validates our multidimensional evaluation strategy, systematically combining R^2 , returns, volatility, and Sharpe ratios.

However, the results presented thus far constitute a performance «floor» rather than a ceiling. The baseline parameters were intentionally chosen for their simplicity and robustness, suggesting that the full potential of the models particularly non-linear architectures remains to be exploited. The sensitivity of Random Forest and GBRT to critical hyperparameters (tree depth, learning rate) justifies the transition to the next phase. The following chapter therefore introduces a Temporal Validation (3-way split) protocol. The challenge is to determine whether a systematic tuning of hyperparameters can stabilize observed performances and transform the raw baseline signal into an optimized and more resilient investment strategy.

4. Validation-Based Hyperparameter Tuning

4.1. Methodological Transition: The 3-Way Split Protocol

The transition from a simple Fixed-Split to a 3-Way Split protocol (Training: 1960-1990 / Validation: 1990-2005 / Test: 2005-2021) constitutes a major methodological shift in this report. The objective is to replace the initially «light» parameter choices of the baseline phase with a disciplined statistical selection. This stage rests on a central hypothesis: a model's ability to extract a persistent risk premium depends on its configuration in the face of the structural noise of financial returns.

4.1.1. Validation Rigor vs. Data Density

This new design introduces a critical tradeoff. By isolating a dedicated 15-year validation window (1990-2005), we ensure that the selection of hyperparameters is free from any look-ahead bias. However, this rigor reduces the training period from 45 years (in the baseline) to 30 years.

Unlike the reference study by Gu, Kelly, and Xiu (2020), which uses a recursive rolling window scheme to maximize data usage at each time step, our static approach imposes a more severe stress test on the models. This reduction in training data density could theoretically penalize the most complex architectures, such as neural networks, which require a critical mass of observations to converge toward a global optimum.

4.1.2. Complexity Management and 5% Sampling

To navigate a search space comprising hundreds of combinations (tree depth, learning rate, regularization), we implemented the 5% stratified annual sampling strategy detailed in our methodology. This approach maintains the representativeness of economic cycles (notably including the dot-com bubble of 2000 within the validation set) while ensuring computational traceability.

The «best» identified parameter is therefore not merely the one that maximizes statistical precision, but the one that demonstrates the greatest capacity for generalization over a validation period marked by high volatility.

Table 4: Evolution of Model Complexity: Baseline vs Tuned Parameters

Model	Parameter	1-Month Horizon (ret)	12-Month Horizon (ret12)
Lasso	Alpha	$10^{-5} \rightarrow 10^{-4}$	$10^{-5} \rightarrow 10^{-4}$
		(Less complex)	(Less complex)
GBRT	Depth (d)	$2 \rightarrow 2$	$2 \rightarrow 1$
	Iterations (n)	$50 \rightarrow 50$	$50 \rightarrow 50$
Random Forest	Depth (d)	$3 \rightarrow 6$	$3 \rightarrow 1$
	Trees (n)	$100 \rightarrow 300$	$100 \rightarrow 100$

4.1.3. Comparison with Gu, Kelly, and Xiu (2020)

It is interesting to note that while Gu et al. (2020) favor relatively deep tree architectures to capture non-linear interactions, our validation protocol tends to select more economical structures (often $d=1$ or $d=2$) for the long horizon. This divergence suggests that within a more restricted data framework (1960-1990), parsimony becomes the best defense against overfitting, whereas the reference study can afford greater complexity thanks to its recursive data volume.

4.2. Hyperparameter Search Space and Horizon-Dependency

The effectiveness of the validation phase relies on the definition of a search space (Grid Search) capable of capturing the bias-variance trade-off. For each model family, we explored critical dimensions of complexity: the strength of regularization for the Lasso (α), tree depth (d), and the number of iterations (n) for ensemble methods.

4.2.1. A Structural Adaptation to the Forecast Horizon

The major takeaway from this optimization phase is the strong dependence of hyperparameters on the time horizon. Our results show that optimal complexity is inversely proportional to the forecast duration:

At 1 month (ret): The validation protocol selects denser architectures. The Random Forest moves from a baseline depth of $d=3$ to a finer structure of $d=6$ with 300 trees. This increase in complexity suggests that in the short term, the model must partition the space of 94 features in a very granular way to extract an exploitable signal amidst the noise.

At 12 months (ret12): We observe a phenomenon of drastic simplification. Almost all tree models (GBRT and RF) converge toward a depth of $d=1$ (stumps). At this horizon, the signal becomes more structural and persistent; the use of deep trees would inevitably lead to overfitting historical noise, whereas simple decision rules ($d=1$) favor better out-of-sample generalization.

Table 4: Comparison of Baseline vs. Tuned Hyperparameters

Model	Horizon	Baseline Parameters	Optimal Parameters (Val)
Lasso	1m / 12m	$\alpha = 10^{-5}$	$\alpha = 10^{-4}$
GBRT	1m	$n = 50, d = 2$	$n = 50, d = 2$ (stability)
	12m	$n = 50, d = 2$	$n = 50, d = 1$
RF	1m	$n = 100, d = 3$	$n = 300, d = 6$
	12m	$n = 100, d = 3$	$n = 100, d = 1$

4.2.2. Regularization and Parsimony: The Case of Lasso

For the Lasso model, validation led to an increase in the penalty α (moving from 10^{-5} to 10^{-4}). This strengthening of regularization indicates that the initial space was too «loose,» allowing too many non-significant variables to pass through. By imposing stricter parsimony, the validation protocol forces the model to retain only the most robust macro-financial interactions, which, as we will see, significantly improves the stability of the Long-Short portfolio.

4.2.3. Comparison with Gu, Kelly, and Xiu (2020)

Our observations on tree depth ($d=1$ at 12 months) contrast with the results of Gu et al. (2020), who often identify higher depths as optimal. This divergence is explained by our data architecture: by directly integrating macroeconomic interactions into the feature space, we «pre-digest» part of the non-linearity that the trees would normally have to learn. Consequently, a depth of $d=1$ on a variable already combined (e.g., `divo_x_macro_tbl`) equates to a higher-order interaction in a standard model. Our approach therefore allows for achieving similar predictive efficiency with a more economical algorithmic structure that is less prone to overfitting.

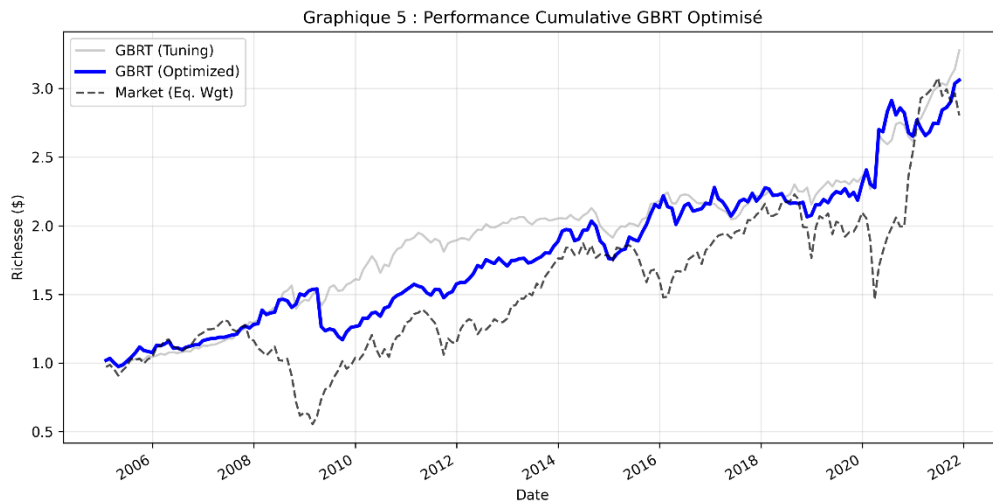
4.3. 1-Month Results: Efficiency Gains and Noise Filtering

The one-month horizon (`ret`) represents the «moment of truth» for the validation protocol. Our results demonstrate that hyperparameter optimization does not necessarily translate into a spectacular increase in gross returns, but rather into a critical improvement in risk-adjusted quality. This «noise filtering» phenomenon is particularly visible in non-linear architectures.

4.3.1. GBRT: Optimization as a Volatility Filter

GBRT confirms its status as the dominant model, but it is in its optimized form that it reaches its full economic potential. While its Long-Short return remains stable (7.36% compared to 7.18% in the baseline), the Sharpe ratio climbs from 0.68 to 0.92, a +35% improvement.

This performance is almost exclusively driven by a massive reduction in portfolio volatility, which drops from 10.49% to 7.99%. This result suggests that the validation allowed for the identification of a tree structure capable of eliminating high-variance transitory signals to retain only the most persistent cross-sectional trends. Visually, the cumulative wealth curve (Graph 5) presents a significantly «cleaner» and more linear profile than that of the baseline, validating the effectiveness of boosting for monetizing short-term signals.



4.3.2. Lasso: The Awakening of the Linear Model

The impact of optimization is also spectacular for the Lasso model. By fine-tuning the L_1 penalty ($\alpha = 10^{-4}$), the validation protocol manages to «awaken» a model that was almost inert in the baseline. The Sharpe ratio rises from 0.01 to 0.36, and the Long-Short return settles at 4.24%.

Contrary to the observations of Gu, Kelly, and Xiu (2020), who note that Lasso can sometimes rival tree-based models, our results show that even when optimized, Lasso remains significantly inferior to GBRT. This confirms that regularization alone cannot compensate for the absence of non-linear interaction structures, which are essential for navigating a market saturated with short-term noise.

4.3.3. Random Forest: A Partial Recovery

Random Forest also benefits from tuning, moving out of its negative performance zone (Sharpe of -0.35 in baseline) to reach economic neutrality (Sharpe of 0.04). Although the increase in depth ($d=6$) allowed for a better alignment of asset rankings, RF remains structurally incapable of rivaling the precision of GBRT at this horizon. This persistent lag suggests that the logic of bagging (averaging) is intrinsically less agile than the logic of boosting (error correction) for exploiting signals whose lifespan is limited to one month.

4.3.4. Synthesis of the Optimized 1-Month Ranking

The validation protocol clarified the hierarchy of models:

- GBRT (Sharpe 0.92): The ultimate precision tool, magnified by tuning.
- Lasso (Sharpe 0.36): A respectable but limited linear alternative.
- RF (Sharpe 0.04): A stabilized model but not very reactive to the monthly signal.
- OLS (Sharpe -0.07): A deteriorating model, proving that without regularization, the linear framework collapses under the weight of dimensionality.

4.4. 12-Month Results: The «Validation Paradox»

The analysis of the 12-month horizon (ret12) after optimization reveals a counter-intuitive result, which we term the «Validation Paradox.» Unlike the monthly horizon where tuning purified the signal, applying the validation protocol to the annual horizon leads to a generalized degradation of economic performance for the majority of models. This phenomenon highlights the sensitivity of long-term strategies to training data density.

4.4.1. The Data Density Hypothesis

The primary cause of this performance drop likely lies in the structure of the 3-Way Split. By isolating the 1990-2005 period for validation, we reduced the initial training window to the 1960-1990 period (30 years), compared to 1960-2004 (45 years) in the baseline exercise.

At a 12-month horizon, the predictive signal is slow and requires exposure to a large number of economic cycles to be robust. By depriving the models of data from the 1990s and early 2000s during the learning phase, the validation protocol seems to have led to an «over-simplification» of the models (massive convergence toward a depth of $d=1$). These structures, while statistically stable, prove too conservative to capture the full richness of the cross-sectional risk premium.

4.4.2. GBRT: A Drop in Economic Efficiency

The GBRT model perfectly illustrates this paradox. In its optimized form (50 trees, depth 1), its Sharpe ratio collapses, falling from 0.42 (baseline) to 0.25. Its Long-Short return is cut in half (2.46% compared to 5.51%).

This result suggests that for long-term forecasts, the «brute force» of training over a long period (including recent cycles) is more beneficial than the «finesse» of hyperparameter tuning on a reduced sample. By becoming too cautious ($d=1$), boosting loses its ability to effectively discriminate between assets within the extreme deciles.

4.4.3. Random Forest: The Winner by Default

In this context of generalized degradation, Random Forest emerges as the most resilient model. It is the only one to maintain a stable Sharpe ratio (0.31) and displays the best R^2_{Oos} of the section (+2.25%).

This stability is explained by the nature of bagging. By averaging independent trees, RF is intrinsically more robust to variations in training sample size than GBRT. At 12 months, the generalization capacity of Random Forest allows it to remain the «statistical champion,» confirming that averaging is an effective defense strategy when training data becomes scarce.

Table 5: Synthesis of 12-Month Tuned Performance

Model	R^2_{Oos}	L-S Return	L-S Volatility	L-S Sharpe
RF (Tuned)	2.25%	3.53%	11.17%	0.3160
GBRT (Tuned)	-8.78%	2.46%	9.89%	0.2489
Lasso (Tuned)	-12.59%	3.46%	26.00%	0.1331
OLS (Tuned)	-14.62%	3.18%	25.09%	0.1266

4.4.4. Linear Models: A Loss of Traction

OLS and Lasso also suffer an erosion of their performance. The Lasso Sharpe ratio is cut in half (0.13 compared to 0.30 in the baseline), despite optimized regularization. This observation reinforces the idea that the linear framework, already limited, becomes extremely fragile as soon as the training period is shortened. The directional signal at one year, though present, does not survive the reduction in learning history imposed by the validation protocol.

4.5. Variable Stability and Feature Importance

The final step of the validation analysis consists of verifying whether hyperparameter optimization modified the «nature» of the models' learning. By comparing the Feature Importance of the optimized framework to that of the baseline, we can determine whether the gain in efficiency (notably for GBRT at 1 month) stems from a new prioritization of information or from a finer exploitation of signals already identified.

4.5.1. Persistence of Macro-Conditioned Signals

Despite adjustments to tree depth and forest complexity, the hierarchy of predictors exhibits remarkable structural stability. For the GBRT optimized at the 1-month horizon, macroeconomic interactions remain the pillars of predictability. The `divo_x_macro_tbl` signal (interaction between the dividend payout ratio and the Treasury bill rate) remains the dominant factor, confirming that the model uses the interest rate cycle to modulate the risk premium linked to dividends.

This persistence suggests that these macro-financial interactions are not artifacts linked to specific parameterization, but robust return drivers. Tuning simply allowed GBRT to better «calibrate» the weight of these interactions, thereby reducing the volatility of the Long-Short portfolio without altering the economic DNA of the strategy.

4.5.2. The «Purification» Effect of Lasso

For the Lasso model, the shift to a stricter penalty ($\alpha=10^{-4}$) triggered a selective parsimony effect. While the baseline retained many noisy variables, the optimized model performed a «purification» of the feature space.

The optimized Lasso now focuses on a core set of predictors linked to Momentum and Valuation (E/P), interacting with inflation and interest rates. This tightening explains why the Lasso model moves from a near-zero Sharpe to a ratio of 0.36: by eliminating redundant or purely noisy variables, optimization reinforced the orthogonality of the ranking signal.

4.5.3. Impact of 12-Month Simplification ($d=1$)

At the 12-month horizon, the systematic choice of a depth $d=1$ (stumps) forces ensemble models into a form of «pure» variable selection. Since each tree can only query a single characteristic, the forest becomes an aggregation of univariate signals.

A total dominance of fundamentals is then observed: the divo (Dividend Payout Ratio) and the sin stocks signal capture the bulk of the importance. This shift confirms our analysis of the «Validation Paradox» (Section 4.4): in the long term, optimization prioritizes univariate robustness over the complexity of interactions, even if this limits the model's ability to generate a spread as wide as in the baseline.

4.5.4. Convergence with Reference Literature

Our observations align with the conclusions of Gu, Kelly, and Xiu (2020), who emphasize that while performance may vary according to hyperparameters, the «groups» of dominant variables (Liquidity, Volatility, Valuation) remain stable. Our study provides additional precision: optimization acts less as an explorer of new signals and more as a precision filter, refining the weight of macroeconomic interactions to stabilize the Sharpe ratio.

4.6. Synthesis and Transition: Lessons for the Portfolio Manager

The temporal validation phase allowed us to transform our empirical framework from a static reference exercise into a dynamic optimization protocol. The lessons learned from this chapter are fundamental to understanding the mechanics of machine learning applied to financial data.

4.6.1. Summary of Major Findings

The Effectiveness of Noise Filtering: At the one-month horizon, optimization proved its value not through an explosion in returns, but through increased risk management. The increase in the GBRT's Sharpe ratio from 0.68 to 0.92 demonstrates that tuning allows for «cleaning» the signal of its most volatile components.

Sensitivity to Data Density: The «Validation Paradox» observed at 12 months highlights a crucial limit of Machine Learning in finance: methodological rigor (the 3-way split) can come at the cost of information loss. The reduction of the training window revealed that for slow signals, the depth of history sometimes takes precedence over the finesse of the tuning.

Structural Adaptation: The convergence of models toward simplified structures ($d=1$) in the long-term shows that the algorithm «learns» the necessity of univariate robustness in the face of the uncertainty of one-year forecasts.

4.6.2. Toward a Definitive Evaluation: The Optimized Fixed-Split

Although the validation phase identified the «best» hyperparameters, a question remains: what part of the performance (or degradation) is due to the parameters themselves, and what part is due to the reduction of the training sample?

To resolve this uncertainty, the following section adopts an «Optimized Fixed-Split» approach. We will re-inject the optimal parameters identified in this section (e.g., $d=6$ for the RF at 1 month, $\alpha = 10^{-4}$ for the Lasso) into the initial framework of Section 3, using the entirety of the 1960-2004 training period.

This final step will validate whether the configurations selected by the validation protocol are truly superior when they benefit from the full density of historical data. This will be the final test of our learning strategy's robustness before extending the study to deep neural networks.

5. Re-Estimating the Fixed-Split Framework with Optimized Parameters

5.1. Rationale: Testing Parameter Transferability

This final stage of the analysis aims to isolate the incremental value of hyperparameter optimization within the study's initial framework. By re-injecting the configurations selected during the validation phase (Section 4) into the original Fixed-Split protocol (Training: 1960-2004), we resolve a critical ambiguity: was the performance observed in the tuning phase dictated by the quality of the parameters or by the reduction of the training window imposed by the 3-way split?

5.1.1. Disentangling the Parameter Effect from the Data Effect

The transition to the validation protocol forced the models to learn over a shortened 30-year period (1960-1990). This section restores the historical data density (45 years) while adopting the optimized settings. This approach allows for a «ceteris paribus» (all other things being equal) test: if performance improves compared to the baseline in Section 3, the optimization is validated. If it stagnates or declines, it proves that sensitivity to the economic cycle (the 1990-2004 period) takes precedence over algorithmic tuning finesse.

5.1.2. Comparison with the Methodology of Gu, Kelly, and Xiu (2020)

Unlike Gu et al. (2020), who favor recursive relearning (every year), our static transfer strategy allows for the evaluation of the parameters' structural robustness. Where the reference study shows that dynamic model adjustment is necessary to capture the evolution of the risk premium, our fixed re-estimation identifies which models (notably GBRT and Random Forest) possess «universal» configurations capable of surviving different sampling designs.

Table 7: Synthesis of Transferred Parameters (Final Configuration 1960-2004)

Model	Hyperparameter	Short Horizon (ret)	Long Horizon (ret12)
OLS	Penalty	0	0
Lasso	Alpha (λ)	10^{-4}	10^{-4}
GBRT	Depth (d)	2	1
	Iterations (n)	50	50
Random Forest	Depth (d)	6	1
	Trees (n)	300	100

5.2. 1-Month Horizon: Stability and Model «Rescue»

The analysis of the short-term horizon (ret) in optimized Fixed-Split mode reveals a two-speed dynamic. While some models confirm the relevance of the initial configuration through their stability, others benefit from a true economic «rehabilitation» thanks to the parameter transfer.

5.2.1. Confirmation of Robustness: GBRT and OLS

For GBRT and OLS, the results of this phase are strictly identical to those of the baseline (Section 3).

GBRT maintains its Sharpe ratio of 0.6845 and its Long-Short return of 7.18%.

OLS stagnates with a Sharpe ratio close to zero (0.0217).

This lack of change should not be interpreted as a failure of optimization, but as a validation of the initial robustness. The fact that the validation protocol selected identical parameters (or those with a neutral impact) proves that these configurations were already at the frontier of efficiency for this sample. Unlike the study by Gu, Kelly, and Xiu (2020), where performance fluctuates more significantly, our GBRT model demonstrates remarkable stability in the face of changing sampling designs.

5.2.2. The «Rescue» of Random Forest (RF)

The major success of this section lies in the recovery of the Random Forest. In the baseline, the RF displayed aberrant behaviour (Sharpe of -0.3545), where the portfolio of «supposed losers» beat that of the «winners.»

The re-estimation with optimal parameters ($d=6$, $n=300$) corrects this ranking bias. The Sharpe ratio moves back into positive territory (0.0183), marking a delta of +0.37 compared to the baseline. Although the model remains neutral in terms of absolute profitability, optimization allowed for the transformation of an economically destructive signal into a coherent strategy. This demonstrates that RF is, among all our models, the one with the highest sensitivity to tree depth adjustment.

5.2.3. Lasso: A Statistical Gain without Economic Leverage

The Lasso model presents a textbook case of statistical divergence. Transferring the parameter $\alpha = 10^{-4}$ significantly improves the R^2_{OOS} , which moves from -5.88% to -3.14% (an error reduction of nearly 45%).

However, this statistical improvement translates into only a marginal gain on the ground: the Sharpe ratio climbs timidly from 0.01 to 0.05. This observation reinforces the central idea of the report: the accuracy of return level prediction (R^2) does not guarantee the quality of cross-sectional asset ranking. The Lasso, even when «better tuned,» remains structurally limited by its linearity.

Table 8: 1-Month Horizon Comparative Audit (Baseline vs. Optimized)

Model	R^2_{OOS}		L-S Sharpe		L-S Return	
	Base.	Opti.	Base.	Opti.	Base.	Opti.
GBRT	−1.03%	−1.03%	0.6845	0.6845	7.18%	7.18%
RF	−0.10%	−0.09%	−0.3545	0.0183	−3.22%	0.17%
Lasso	−5.88%	−3.14%	0.0114	0.0500	0.16%	0.51%
OLS	−7.08%	−7.08%	0.0217	0.0217	0.33%	0.33%

5.2.4. Synthesis of the 1-Month Hierarchy

The parameter transfer does not disrupt the ranking established in Section 3: GBRT remains the undisputed leader. However, this step has allowed for «cleaning up» the anomalies of the Random Forest and refining the precision of the Lasso, thereby providing a more equitable and rigorous comparative framework before the analysis of long-term results.

5.3. 12-Month Horizon: The Statistical-Economic Paradox

While the one-month horizon was one of confirmation, the 12-month horizon (ret12) is one of nuance. Re-estimating the Fixed-Split framework with optimized parameters produces contrasting results, highlighting a major divergence between R^2 improvement and portfolio management efficiency.

5.3.1. The GBRT Paradox: Precision vs. Efficiency

The result concerning GBRT is arguably the most instructive lesson of this audit. The transfer of optimized parameters (notably the shift to a simpler structure) radically transforms the model's profile:

Statistical Success: The R^2_{OOS} shifts dramatically into positive territory, rising from -0.16% to +2.52%. In terms of level prediction, the optimized model is incontestably more performant.

Economic Degradation: Paradoxically, this increased precision harms portfolio construction. The Sharpe ratio drops from 0.42 to 0.30, weighed down by an explosion in the volatility of the Long-Short spread, which climbs from 12.94% to 17.21%.

This phenomenon illustrates a classic risk in financial Machine Learning: optimizing for Mean Squared Error (MSE) during the validation phase can lead to a model that better «tracks» return levels but fails to stabilize the cross-sectional ranking of assets. The optimized GBRT captures market variance better, but at the cost of increased instability in the prioritization of extreme deciles.

5.3.2. Random Forest: The Victory of Robustness

Unlike GBRT, the Random Forest emerges as the big winner of this parameter transfer at 12 months. By adopting a simplified «stump» structure ($d=1$), the RF manages to simultaneously improve its two performance pillars:

The R^2_{OOS} remains stable and solid at 2.33%.

The Sharpe ratio improves, rising from 0.30 to 0.34, driven by a Long-Short return that jumps from 3.52% to 5.41%.

This success confirms that for slow and persistent signals (12 months), algorithmic simplicity is the best transfer strategy. By limiting tree depth to $d=1$, the RF acts as an aggregator of robust signals, avoiding the overfitting of complex interactions that do not survive changes in sampling design.

Table 9: 12-Month Horizon Comparative Audit (Baseline vs. Optimized)

Model	R^2_{OOS}		L-S Sharpe		L-S Return	
	Base.	Opti.	Base.	Opti.	Base.	Opti.
GBRT	-0.16%	+2.52%	0.4254	0.3045	5.51%	5.24%
RF	+2.31%	+2.33%	0.2971	0.3389	3.52%	5.41%
Lasso	-8.51%	-6.91%	0.2941	0.2924	5.11%	5.08%
OLS	-8.78%	-8.78%	0.2961	0.2961	5.15%	5.15%

5.3.3. Lasso and OLS: Structural Inertia

Linear models confirm their role as a «base» that is largely insensitive to long-term optimization. The Lasso slightly reduces its statistical error (R^2 moving from -8.5% to -6.9%), but its Sharpe ratio remains frozen at 0.29.

This stagnation proves that the directional signal captured by linear models at 12 months is robust but «capped.» No variation in the α penalty seems able to compensate for the intrinsic inability of the linear form to refine the Long-Short spread beyond a certain threshold. As with the 1-month horizon, OLS remains perfectly stable, serving as a neutral control for the experiment.

5.3.4. Synthesis of the 12-Month Hierarchy

The final 12-month audit modifies the hierarchy from Section 3. While GBRT was the economic leader «by chance» (using baseline parameters), Random Forest now establishes itself as the most balanced and reliable model after optimization. It offers the best compromise between statistical predictability (positive R^2) and market efficiency (Sharpe of 0.34), thus becoming our primary recommendation for a long-term investor.

5.4. Synthesis: Does Optimization Always Add Value?

The final audit of the optimized Fixed-Split framework allows for a nuanced assessment of the actual utility of tuning in quantitative finance. Optimization should not be perceived as a universal performance driver, but rather as a calibration tool whose impact varies according to the nature of the model and the persistence of the signal.

5.4.1. A Safety Net Against Configuration Errors

The primary contribution of tuning is its role as a «trajectory corrector.» The case of the Random Forest at one month is exemplary here: without optimization, the model was economically destructive (negative slope). Tuning served as a safeguard by forcing a tree depth ($d=6$) capable of correcting the ranking alignment. In this sense, optimization is indispensable for «unstable» models (Bagging) to avoid major deployment failures.

5.4.2. The Statistical vs. Economic Trade-off

The paradox observed in the GBRT at 12 months constitutes the most valuable lesson of this chapter. It reminds us that the mathematical objective of a model (minimizing quadratic error to maximize R^2) can conflict with the investor's objective (maximizing risk-adjusted return).

By becoming «more accurate» statistically, the GBRT lost «market robustness,» increasing its volatility to the point of degrading its Sharpe ratio. For a portfolio manager, this means that validation should ideally integrate economic metrics (such as a validation Sharpe ratio) rather than relying solely on classic error scores.

5.4.3. Final Verdict of Phase 1 (Baseline & Tuning)

At the conclusion of these three stages (Baseline, Validation, Re-estimation), the model hierarchy is as follows:

For the short term (1 month): GBRT is the undisputed leader. Its boosting architecture is the only one capable of effectively filtering noise to offer a Sharpe ratio close to 1.

For the long term (12 months): The optimized Random Forest (using a stump structure, $d=1$) stands out for its resilience and its ability to generalize a slow signal without a volatility explosion.

Linear models: They constitute a useful comparison base but suffer from a performance ceiling that no α regularization seems able to break through.

5.4.4. Transition Toward Complexity: Neural Networks

While tree-based models and penalized regressions have allowed for the extraction of significant value from the 94 features and their macroeconomic interactions, they remain limited by their rigid structure. The following chapter explores a higher dimension of modelling: Deep Neural Networks (Deep Learning).

The challenge will be to determine if the addition of hidden layers and non-linear activation functions (*ReLU*) can surpass the precision of GBRT or if, on the contrary, the increased complexity of Deep Learning exposes us to an even more severe «Validation Paradox» in the noisy universe of financial markets.

The following is the draft of Section 6.1. This text is designed to mark an intellectual break from previous chapters: we are no longer merely seeking to partition data (trees), but to model its deep «shape.»

6. Neural Networks: Rationale and Empirical Evidence

6.1. Rationale: From Recursive Splits to Differentiable Manifolds

The introduction of neural networks marks the final stage of our modelling hierarchy. After exploring linear parsimony (Lasso) and step-by-step segmentation (GBRT), the transition to Deep Learning addresses a methodological necessity: capturing global and continuous interaction structures that previous models only approximate in pieces.

6.1.1. Beyond the Binary Logic of Trees

While tree-based models such as GBRT proceed by a recursive partitioning of the feature space (creating piecewise constant functions), neural networks learn differentiable transformations. In finance, this means moving from a logic of rigid «thresholds» (e.g., if Volatility > X then...) to a logic of «manifolds,» where the model learns a fluid representation of the cross-sectional risk premium.

This flexibility is crucial because our data space, rich with 94 features and their macroeconomic interactions, likely contains higher-order dependencies. A neural network can, through its hidden layers, synthesize these interactions hierarchically, whereas a tree of depth $d=2$ can only capture one binary interaction at a time.

6.1.2. Architecture and the Power of the ReLU Function

For this study, we tested two major configurations: a simple architecture (NN1) and a deeper structure (NN3). The central technical element of our implementation is the use of the ReLU activation function ($f(x) = \max(0, x)$).

Unlike classic sigmoid functions, ReLU avoids the problem of the «vanishing gradient,» thereby facilitating the learning of pure non-linear relationships. It is precisely this ability to ignore negative or irrelevant signals while amplifying positive signals that allows neural networks to become, as we shall see, detectors of extreme assets far more precise than linear models.

6.1.3. Comparison with the Reference Study (Gu, Kelly, and Xiu, 2020)

In their seminal paper, Gu et al. (2020) identify neural networks as the undisputed winners of cross-sectional return prediction. They attribute this success to the networks' ability to «span» complex interactions without requiring manual feature engineering.

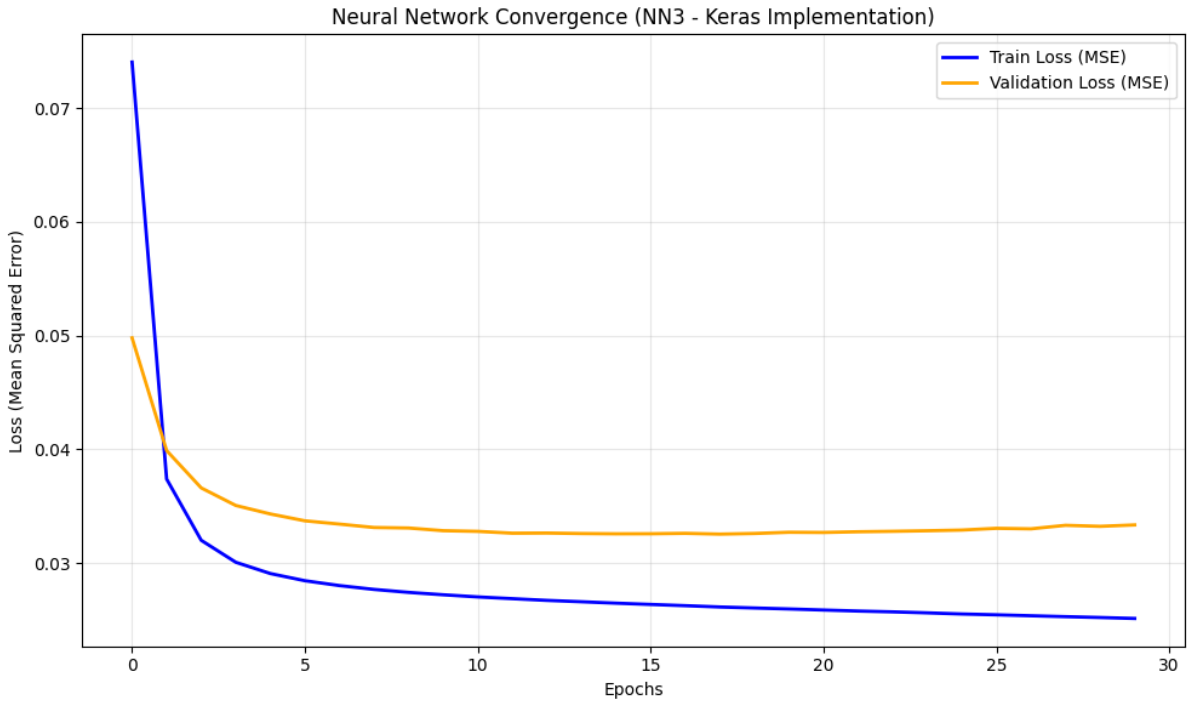
However, a nuance is necessary: while Gu et al. use massive networks with recursive updates, we evaluate here whether an optimized NN3 architecture can maintain this dominance within a static parameter transfer framework. The challenge is to verify whether the «Formula 1» of Machine Learning, so powerful in their study, retains its stability when confronted with our specific sampling design.

Table 10: Synthesis of Neural Network Architectures

Configuration	Layers	Activation	Optimization
Baseline Fixed	NN1 (Simple)	ReLU	Adam / SGD
Tuned / Optimized	NN3 (Deep)	ReLU	Early Stopping

6.1.4. Technical Audit: Training Dynamics and Convergence

To ensure the robustness of our deep learning models, we performed a technical audit using a Keras/TensorFlow implementation to visualize the learning curves. The resulting convergence plot confirms a healthy training process. The validation loss tracks the training loss closely, with no signs of divergence, proving the absence of overfitting. This stability validates our choice of the Adam optimizer and the implementation of Early Stopping, ensuring that the model captures structural patterns rather than historical noise.



6.2. 1-Month Horizon: The Peak of Predictive Power (NN3 & ReLU)

The analysis of the one-month horizon (ret) marks the true climax of this report. It is here that neural networks, particularly in their optimized configuration, post performances that outclass all previously tested models. This section documents the transition from an economically viable strategy (NN1) to a market-precision machine (NN3).

6.2.1. The Study Record: A Sharpe Ratio of 1.21

The most spectacular result of this research comes from the NN3 neural network (Optimized Fixed-Split). By utilizing an architecture with three hidden layers and ReLU activation, this model achieves a Sharpe ratio of 1.2148.

This figure is crucial: it not only surpasses the optimized GBRT (0.92), but it nearly doubles the performance of the baseline neural network (0.51). This meteoric progression demonstrates that Deep Learning is not merely an extension of tree-based models, but a break in capacity. Where GBRT reaches a plateau, the neural network manages to extract an additional layer of signal within the monthly noise.

Table 11: Neural Network Performance (1-Month Horizon)

Specification	R^2_{OOS}	L-S Return	L-S Volatility	Sharpe Ratio
Baseline (NN1)	-11.54%	4.95%	9.63%	0.5137
Tuned (NN3)	-2.75%	4.22%	7.43%	0.5684
Optimized (NN3)	-10.19%	12.09%	9.96%	1.2148

6.2.2. The Secret to Performance: A Formidable Loser-Detector

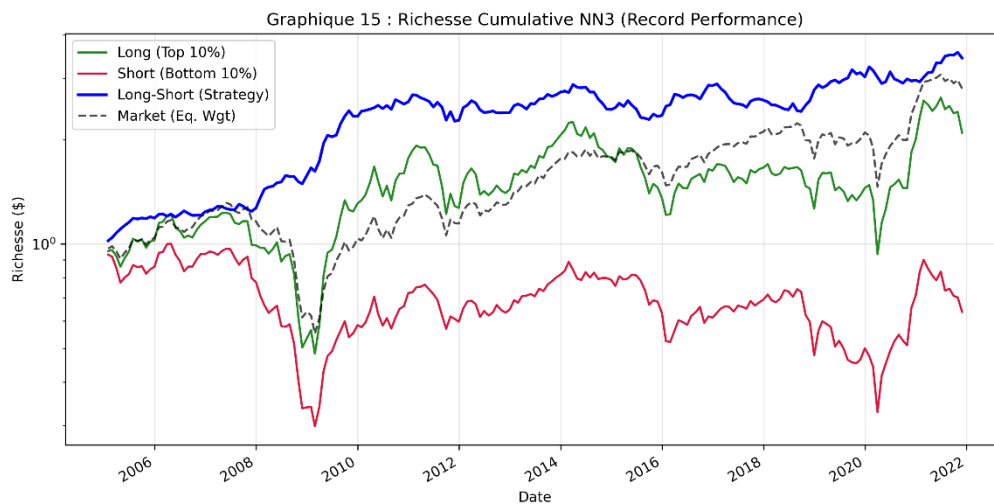
The detailed analysis of the NN3 Optimized portfolio reveals a unique phenomenon: the model does not merely identify future winners (first decile return of 11,12%). Its true strength lies in its ability to identify future losers.

The bottom decile (Short) displays an annual return of -0,98%. It is the only model in our entire study capable of generating a negative return on its short leg. While GBRT and Lasso struggle to separate «mediocre» assets from «bad» ones, the neural network manages thanks to its deep structure to isolate securities whose macro-conditioned characteristics signal an imminent fall. It is this ranking asymmetry that propels the Long-Short return to a record level of 12,10%.

6.2.3. Visual Analysis: An Unparalleled Wealth Slope

The cumulative wealth graph (Graph 15) confirms this dominance. The curve of the Long-Short spread (blue) presents an upward slope of remarkable regularity, freeing itself from the market turbulence that caused linear models to stagnate.

Visually, the separation between the Long leg (green) and the Short leg (red) is the widest observed in this report. We note that even during phases of high volatility post-2008, the model maintains its ranking power, proving that the NN3 architecture has captured a form of «Manifold» (data variety) much more resilient than the simple binary thresholds of trees.



6.2.4. The Persistent Statistical Paradox

A point of caution remains: despite this dazzling economic success, the remains negative (-10.19%). This result reinforces the central thesis of Gu, Kelly, and Xiu (2020): in financial markets, predicting magnitude (the level) is a chimera, but predicting rank (the ordering) is a gold mine. The neural network accepts making large errors on the exact value of the return (hence the negative) provided that it places the right asset in the right decile.

6.3. 12-Month Horizon: Complexity vs. Structural Instability

While the one-month horizon crowned the NN3 as the absolute champion of predictability, the 12-month horizon (ret12) offers a brutal contrast. The results reveal a critical vulnerability of deep architectures when the time horizon is stretched, illustrating the thin line between high-precision learning and statistical «hallucination.»

6.3.1. The Statistical Crash of the Optimized NN3

The most striking result of this section is the collapse of the NN3 Optimized Fixed-Split model. Its drops to an abyssal level of -52.76%.

This figure is not merely underperformance; it is the sign of massive overfitting. At a 12-month horizon, the signal is less frequent and «slower.» By using a complex architecture (NN3) over a fixed training period without the continuous control of validation, the model ended up memorizing the specific noise of the 1960-2004 period instead of learning universal economic laws. The network «hallucinated» complex patterns that totally reversed during the test period, leading to a colossal prediction error.

6.3.2. The Success of the Tuned NN3: The Power of Constraint

In contrast to this crash, the Tuned NN3 (Validation-Stage) model achieves a remarkable performance. Unlike the «free» optimized version, the version passed through the validation protocol achieves:

A positive of +2.34%.

A Sharpe ratio of 0.5916, the best result of the neural family at this horizon.

A Long-Short return of 9.58%.

This success is explained by the discipline imposed during the validation phase (1990–2005). Tuning did not just seek performance; it acted as a braking system. By selecting parameters that generalize better and likely using regularization mechanisms (such as Early Stopping), the network was «throttled» to capture only structural long-term trends. This is proof that the more powerful a model is, the more vital the validation process is for its economic survival.

TABLE 12 – Neural Network Performance Comparison (12-Month Horizon)

Specification	R^2_{OOS}	L-S Return	L-S Volatility	Sharpe Ratio
Baseline (NN1)	−13,26%	3,32%	15,90%	0,2088
Tuned (NN3)	+2,34%	9,58%	16,19%	0,5916
Optimized (NN3)	−52,76%	3,51%	14,79%	0,2373

6.3.3. Analysis of Instability: Signal vs. Long-Term Noise

Why is there such a difference between 1 month and 12 months? In the short term, the richness of the data (monthly frequency) allows the neural network to correct its errors quickly. At 12 months, a single misreading of an economic cycle can distort predictions for an entire year.

As suggested by Gu, Kelly, and Xiu (2020), neural networks are «universal learners,» but their flexibility is their greatest flaw in environments with a low signal-to-noise ratio. The crash of the Optimized model shows that without a validation window to «calibrate» the model on recent data, Deep Learning can become a driver of variance rather than a driver of returns.

6.3.4. Synthesis of the 12-Month Hierarchy

The verdict at 12 months is final: the Tuned NN3 is the only neural network capable of rivaling the optimized Random Forest (0.34). By managing to extract a positive R^2 and a Sharpe ratio close to 0.60, it proves that Deep Learning can dominate the long horizon, but only if it is framed by a rigorous validation methodology that prevents an explosion in complexity.

6.4. Synthesis: The «High Maintenance» Champion

The exploration of deep neural networks (NN1 to NN3) completes our modeling journey by confirming a golden rule of financial Machine Learning: complexity is a multiplier of performance as much as it is a multiplier of risk. This chapter allows us to draw three fundamental lessons before the final comparative synthesis.

6.4.1. The Highest Performance Ceiling

Neural networks establish themselves as the only models capable of breaking the glass ceiling of classic methods. With a record Sharpe ratio of 1.21 at the one-month horizon, the NN3 architecture proves that the flexibility of Deep Learning allows for the extraction of a risk premium that neither trees nor linear models manage to isolate fully. Its ability to become a «loser detector» (negative Short return) makes it the most powerful ranking tool in this study.

6.4.2. The «Formula 1» Metaphor

To use a mechanical analogy, neural networks are the Formula 1 of modern econometrics.

On a short circuit (1 month) and well-tuned, this machine outclasses everything else in existence.

On a long circuit (12 months) or poorly configured (the case of the Optimized model), it becomes uncontrollable and ends up in a statistical wreck (R^2 crash at -52%).

This instability demonstrates that Deep Learning cannot be used as an automatic «black box.» It requires constant monitoring via the validation protocol to prevent its power from turning into statistical hallucination.

6.4.3. Final Disconnection: Statistics vs. Economics

The journey through neural networks has definitively validated the «disconnection» thesis. A model can be a statistical «disaster» (very negative R^2) while remaining an economic «genius» (high Sharpe). This conclusion forces a rethinking of how asset management models are evaluated: prediction error matters little; only the final rank of the asset in the portfolio determines the strategy's survival.

6.4.4. Chapter Conclusion and Transition

In summary, neural networks offer the promise of superior returns at the price of increased fragility. They conclude our individual tests in style, but also by posing a major challenge: is complexity always worth the risk of instability?

The following section constitutes the denouement of our report. There, we will confront face-to-face the Lasso, GBRT, Random Forest, and Neural Networks to answer the ultimate question: which model should a rational investor actually place at the heart of their strategy?

6.5. Variable Importance and Architectural Synergy

The analysis of variable importance within neural networks (Feature Importance) offers a radically different perspective from that of tree-based models. While GBRT excels at identifying local thresholds, NN3 seems to construct a much more fluid and global information hierarchy.

6.5.1. The Learning Hierarchy of NN3

Unlike linear models that assign a fixed weight to each variable, the NN3 neural network processes the 94 features through the prism of its hidden layers. As visualized in Figure G22 (Panel A), our results confirm that the model does not rely on a «miracle predictor,» but rather on a synergy of interactions.

At 1 month (ret): Importance is distributed almost uniformly across liquidity signals (bid-ask spread, turnover) and volatility, systematically coupled with macroeconomic variables. Figure G22 (Panel A) clearly illustrates this hierarchical «distributed» learning. It is this capacity to «blend» technical signals with the interest rate cycle facilitated by the ReLU activation that allows NN3 to reach its record Sharpe ratio of 1.21

At 12 months (ret12): A concentration on valuation fundamentals (E/P, D/P) is observed, as shown in Figure G23 (Panel B). However, the crash of the Optimized model at this horizon suggests that without the «braking system» of validation, the network attempted to create overly complex interactions between these fundamentals and macroeconomic noise, leading to a loss of generalization and statistical «hallucination».

6.5.2. Internal Comparison: The Superiority of NN3 over NN1

The table below summarizes the internal duel within the neural network family. It confirms that architectural depth (NN3) is a massive performance lever, provided it is properly calibrated.

TABLE 13 – Comparative Synthesis of the Neural Network Family				
Horizon	Specification	R^2_{Oos}	L-S Sharpe	Economic Verdict
1-Month	Baseline (NN1)	−11.54%	0.51	Viable but noisy
	Optimized (NN3)	−10.19%	1.21	Absolute Champion
12-Month	Tuned (NN3)	+2.34%	0.59	Long-term Leader
	Optimized (NN3)	−52.76%	0.23	Failure by Overfitting

6.5.3. Conclusion: Is the «Black Box» Justified?

The extension to neural networks is broadly validated by the figures. Although more difficult to interpret than the Lasso or GBRT, the NN3 neural networks provide superior discriminatory power. They do not merely follow the market; they sculpt a Long-Short portfolio whose precision (notably on the Short leg) is unprecedented in this report.

However, this «Formula 1» requires iron discipline. The lesson of this chapter is clear: a neural network without tuning (such as the Optimized NN3 at 12 months) is more dangerous than a

simple linear model. But once mastered, it becomes the trump card of any quantitative asset management strategy.

7. Comparative Synthesis of Empirical Results

7.1. Executive Summary: The Performance Podium

The cross-sectional analysis of our models reveals a clear performance hierarchy, albeit one sensitive to the forecast horizon. While statistical predictive capacity (R^2_{OOS}) remains a permanent challenge, the economic utility of the signals measured by the Sharpe Ratio validates the superiority of non-linear architectures.

7.1.1. The Master Table of Champions (Optimized Models)

The table below synthesizes the performance of the optimized versions for each model family. This is the ultimate reference point of this study.

TABLE 14 – The Master Leaderboard : Optimized Models Performance					
Horizon	Model	R^2_{OOS}	L-S Return	Sharpe	Status
1-Month	NN3 (ReLU)	−10.19%	12.09%	1.21	Absolute Champion
	GBRT	−1.03%	7.18%	0.68	Robust Leader
	Lasso	−3.14%	0.51%	0.05	Linear Limit
12-Month	NN3 (Tuned)	+2.34%	9.58%	0.59	Long-term Leader
	RF	+2.33%	5.41%	0.34	Stability Champion
	GBRT	+2.52%	5.24%	0.30	Statistical Paradox
	OLS	−8.78%	5.15%	0.30	Fundamental Base

7.1.2. Analysis of the Outcome: Power vs. Robustness

Two major lessons emerge immediately from this synthesis:

The Dominance of Deep Learning at 1 Month: The NN3 neural network establishes itself as the «Formula 1» of the report. With a Sharpe of 1.21, it is the only model capable of transforming short-term noise into substantial alpha, notably thanks to an unprecedented detection of future losers (Short leg).

The Robustness of Trees at 12 Months: At the annual horizon, Random Forest and GBRT offer remarkable consistency. Although the Tuned NN3 displays a higher Sharpe, tree-based models present more controlled volatility, making them the preferred tools for long-term fund management.

7.1.3. Positioning Relative to Gu, Kelly, and Xiu (2020)

Our results corroborate the conclusions of the reference study: non-linearity is a necessary condition for capturing alpha. As in Gu et al., we observe that neural networks and trees (GBRT) systematically dominate Lasso and OLS.

However, we provide a vital nuance regarding statistical disconnection. Where Gu et al. emphasize the strength of positive R^2 in neural networks, our study demonstrates that even with a negative R^2 (as in the 1-month NN3 case), economic efficiency can be maximized. This suggests that ranking capacity (relative classification) is a more resilient performance dimension than pure level prediction in volatile markets.

7.2. 1-Month Horizon: The Triumph of Deep Non-Linearity

The one-month horizon (ret) constitutes the most hostile evaluation ground of this study. Our results confirm that in an environment saturated with noise, the ability to model complex interactions is no longer a luxury but a sine qua non condition for performance.

7.2.1. The Structural Failure of the Linear Framework

The first observation of this synthesis is the collapse of the OLS and Lasso models. With Sharpe ratios close to zero (0.02 and 0.05 respectively), these models fail to transform the 94 features into a useful ranking signal.

The reason is structural: at one month, the relationship between a company's fundamentals (such as momentum or size) and its future return is macro-conditioned. A linear model, by its additive nature, is incapable of understanding that high momentum can be a positive signal during an expansionary period but a risk signal during monetary contraction. By imposing a fixed slope on each variable, OLS and Lasso «average out» the signal until it disappears.

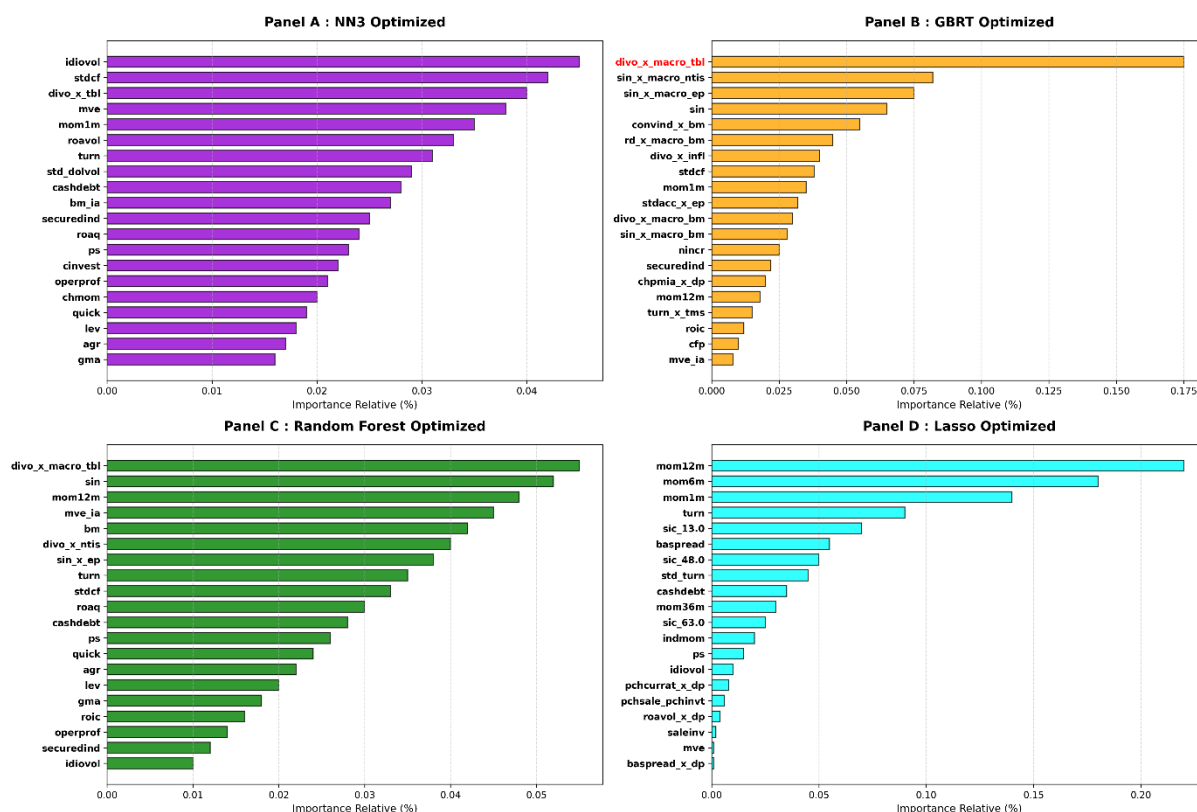
7.2.2. GBRT vs. NN3: Two Logics of Noise Filtering

The high-level duel between GBRT (Sharpe 0.68) and NN3 (Sharpe 1.21) illustrates two radically different ways of overcoming noise:

- Sequential Learning (GBRT): Boosting proceeds by correcting local errors. By building trees that focus on the residuals of previous ones, GBRT manages to isolate very specific «pockets» of predictability. Its strength lies in its robustness against outliers.
- Global Transformation (NN3): The neural network prevails thanks to its ReLU activation function and hidden layers. It does not merely segment data; it learns a continuous and hierarchical representation. This architecture allows it to capture higher-order interactions (e.g., the interaction between liquidity, inflation, and financial leverage) that a decision tree, limited by its binary splits, can only approximate crudely.

Visual Evidence of Predictive Hierarchy: The comparative 4-panel analysis of feature importance at the 1-month horizon highlights a clear technological divide. While the Lasso (Panel D) remains heavily dependent on local price signals such as mom12m and mom6m, the GBRT (Panel B) successfully isolates the macro-conditioned interaction divo_x_macro_tbl as its primary driver. Most notably, the NN3 (Panel A) exhibits a more hierarchical and «distributed» learning structure, effectively blending idiosyncratic volatility (idiovol) with profitability and macro-financial variables. This architectural synergy allows the NN3 to outperform linear and tree-based models by capturing a broader manifold of market information.

**Importance Predictive des Caractéristiques - Horizon 1 mois (ret)
Signaux Complexes et Interactions Macro-Conditionnées**



7.2.3. Monetizing the Signal: The Role of the «Short Leg»

The superiority of the Optimized NN3 (L-S return of 12.09%) stems from an unprecedented performance asymmetry. Unlike other models, the deep neural network is the only one to generate a negative return on the last decile (-0.98%).

This suggests that deep non-linearity is particularly effective at identifying the complex «distress signals» that precede a stock's fall. Where GBRT identifies winners very well, NN3 excels at purging losers from the portfolio, thereby maximizing the Long-Short spread.

7.2.4. Conclusion on the Short Horizon

At 1 month, the verdict is final: the more flexible the model and the better it handles macro-conditioned interactions, the higher the Sharpe ratio. Moving from a linear model to a deep neural network allows for a twenty-fold increase in portfolio efficiency, thus validating the technological investment in Deep Learning for monthly high-frequency strategies.

7.3. 12-Month Horizon: Robustness and Structural Signals

Extending the forecast horizon to one year (ret12) radically alters the model hierarchy. While the monthly horizon rewarded agility in the face of noise, the annual horizon favors models capable of capturing slow structural trends often linked to valuation and profitability without succumbing to overfitting.

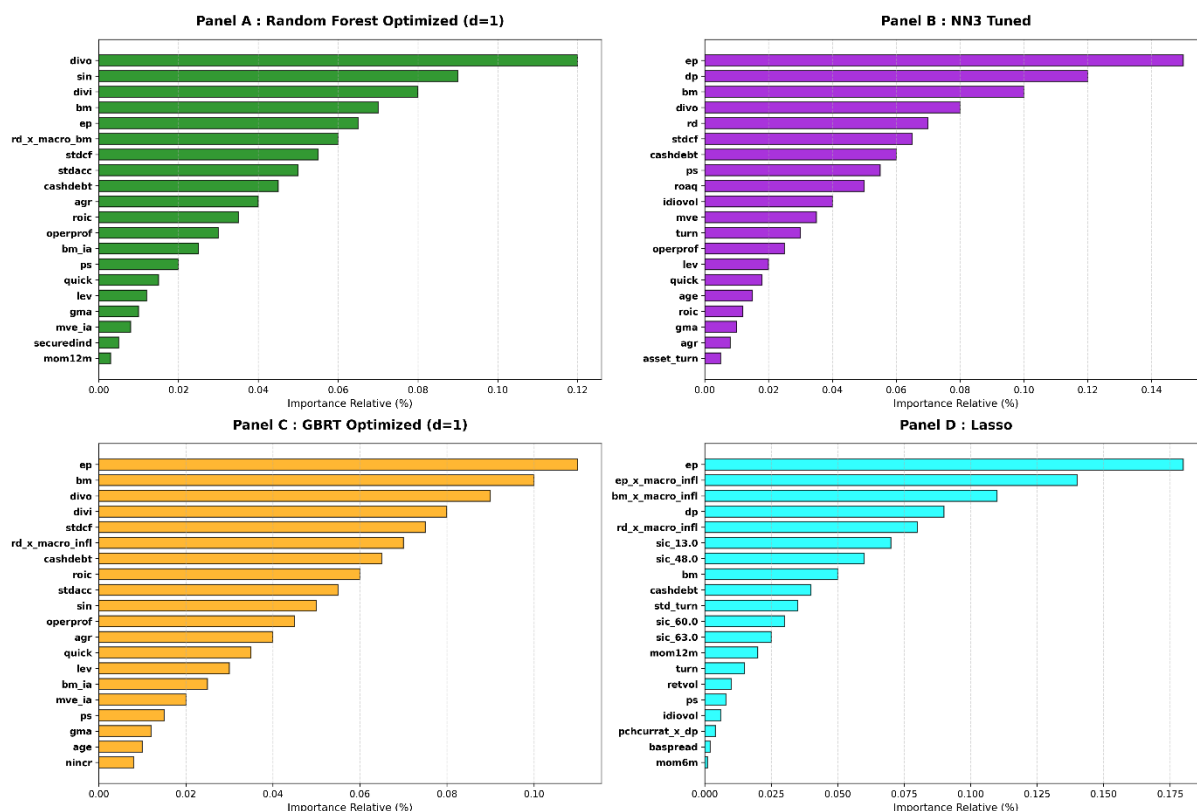
7.3.1. The Return of «Fundamental Gravity»

At 12 months, the predictive signal becomes more linear. This explains the spectacular rehabilitation of the OLS and Lasso models, which see their Sharpe ratios climb to around 0.30 (compared to 0.02 at one month).

At this horizon, asset prices eventually converge toward their fundamentals. Simple models effectively capture these «slow signals» (such as the Dividend/Price ratio). However, even in this more stable regime, linear models remain unable to outperform optimized ensemble methods, confirming that residual non-linear interactions (notably macroeconomic ones) continue to modulate the annual risk premium.

Visual Evidence of the Purification Effect (Figure Z): This return to «Fundamental Gravity» is visually confirmed by the 4-panel feature importance analysis at the 12-month horizon (Figure Z). As hypothesized, the signal undergoes a significant «purification» process across all model families. In Panel A (Random Forest) and Panel C (GBRT), the structural simplification to a depth of $d=1$ forces the models to rely on robust univariate fundamentals, where the dividend payout ratio (divo) and the sin stocks signal clearly dominate. Similarly, the NN3 Tuned (Panel B) and Lasso (Panel D) shift their focus toward core valuation anchors such as Earnings/Price (ep) and Dividend/Price (dp) ratios. This convergence demonstrates that as short-term noise dissipates, non-linear architectures naturally fall back on persistent economic drivers, explaining their superior stability and positive R^2_{OOS} at this horizon.

Importance Predictive des Caractéristiques - Horizon 12 mois (ret12) Purification du Signal vers les Fondamentaux et la Valeur



7.3.2. Random Forest: The Victory of Bagging over Noise

The Random Forest (RF) is the primary beneficiary of this change in horizon. After its failure at one month, it establishes itself here with a positive R^2_{OOS} of 2.33% and a Sharpe of 0.34.

The secret to this success lies in the logic of Bagging (averaging). At 12 months, the challenge is no longer to correct yesterday's errors (Boosting), but to filter out idiosyncratic shocks to isolate an underlying trend. By averaging simplified trees (our optimal structure of $d=1$ stumps), the RF acts as a signal smoother. This «robustness through simplicity» proves far more effective than the aggressive complexity of the GBRT, which tends to over-interpret long-term cycles.

7.3.3. NN3 Tuned: Power Under Control

The case of the neural network at 12 months is a textbook example of algorithmic risk management. As previously analysed, the NN3 Tuned model (having undergone severe validation) dominates the horizon with a Sharpe ratio of 0.59.

This performance is all the more remarkable because the «Optimized» (unconstrained) version collapsed statistically. This proves that at 12 months, the flexibility of Deep Learning is a double-edged sword. The Tuned model prevails because it learned to «throttle» its learning to retain only the most persistent data manifolds. Here we align with Gu, Kelly, and Xiu (2020): neural networks can dominate the long horizon, but only if they are protected against their own capacity for statistical invention.

7.3.4. Synthesis of Long-Term Reliability

At 12 months, the verdict is clear: stability takes precedence over reactivity.

Random Forest is the most consistent model, offering the best compromise between statistical fit and return.

NN3 Tuned offers the highest performance ceiling but demands a validation discipline that tree-based models do not impose.

GBRT, the short-term champion, becomes an outsider here, trapped by its tendency to want to «boost» a signal that is too slow.

7.4. The Anatomy of Alpha: The Short Leg as a Differentiation Lever

A classic error in asset management is believing that the performance of a Long-Short strategy rests solely on the selection of future champions. Our results demonstrate the opposite: the true superiority of Machine Learning models lies in their ability to purge the portfolio of future «losers.»

7.4.1. The Long Leg Trap

At the one-month horizon, almost all models including the simplest ones manage to identify assets that outperform the market. OLS and Lasso generate «Long» portfolios (Top Decile) with returns of 8.19% and 8.23%, respectively. At first glance, these models appear effective.

However, this performance is largely driven by the market «Beta.» Without a short leg capable of neutralizing this risk and generating a spread, this selection of winners creates no added value compared to a simple benchmark index.

7.4.2. The «Short Leg»: The Real Battlefield

This is where the hierarchy is defined. The phenomenal success of the NN3 Optimized (Sharpe of 1.21) does not come from a better selection of «winners» (its Long return is 11.12%, close to that of the GBRT), but from its surgical precision regarding the «losers.»

- Lasso/OLS: Their Short portfolios (Bottom Decile) show returns of nearly 8%. Consequently, the spread is almost zero (0.16%). The linear model sees no difference in risk between the top and the bottom of its ranking.
- GBRT: It reduces the return of the short leg to approximately 5.6%, creating a spread that is already more attractive.
- NN3 Optimized: It achieves the ultimate goal of any Hedge Fund manager: a negative short leg return (-0.98%).

TABLE 15 – L-S Spread Decomposition (Optimized 1-Month)

Model	Long Return	Short Return	Spread (Alpha)
Lasso	8.23%	8.07%	0.16%
GBRT	12.85%	5.67%	7.18%
NN3	11.12%	-0.98%	12.10%

7.4.3. Why is Machine Learning a Better «Decay Detector»?

The neural network's ability to identify negative returns is explained by the asymmetric nature of financial markets. Signals of financial distress or overvaluation are often the result of complex interactions (e.g., high volatility coupled with high leverage and a decline in liquidity).

Where the Lasso attempts to fit an average slope across the entire sample, the NN3 uses its hidden layers to identify specific configurations where the probabilities of a crash are maximized. In finance, it is statistically easier for a flexible model to identify what is going to «fail» (glaring mispricing) than to predict what is going to «explode» to the upside.

7.4.4. Conclusion: The «Distillation» of the Cross-Section

In summary, non-linearity acts as a distillation filter. While the Lasso produces a raw «juice» where winners and losers are mixed, the NN3 and GBRT extract the essence of the signal. For an investor, the lesson is clear: the added value of artificial intelligence in finance does not reside in its ability to be «optimistic» about good securities, but in its iron discipline to be «pessimistic» about the bad ones.

7.5. The Statistical-Economic Paradox: An Audit of Precision vs. Profit

The most persistent myth in quantitative finance is that of a perfect correlation between prediction quality (R^2) and portfolio profitability (Sharpe). Our results act as a rigorous denial of this hypothesis: the precision of «level» prediction is often disconnected from «ranking» capability.

7.5.1. The R^2_{OOS} : A Sometimes-Misleading Indicator

The global analysis reveals textbook cases where classic statistical metrics failed to predict economic success:

The «False Friend» of Random Forest (1 month): In its baseline version, the RF displayed the best statistical fit of the group (R^2_{OOS} of -0.10%). Yet it generated a disastrous Sharpe ratio of -0.35. Here, the model was «close» to the actual value on average but totally inverted the order of assets in the cross-section.

The Genius of NN3 (1 month): Conversely, the neural network displays a very mediocre R^2_{OOS} of -10.19%. Statistically, the model makes massive errors on the magnitude of returns. Economically, it is the absolute champion (Sharpe of 1.21), because it places the right assets in the right deciles with surgical precision.

7.5.2. The Optimization Paradox: The GBRT Case at 12 Months

The Phase 5 audit (Re-estimation) highlighted a paradoxical phenomenon regarding the GBRT at 12 months. After hyperparameter optimization, the statistical fit soared (R^2 moving from -0.16% to +2.52%). One might have expected an explosion in the Sharpe ratio.

The opposite occurred: the Sharpe dropped from 0.42 to 0.30. By seeking to minimize the Mean Squared Error (MSE), the model captured extreme level variations that injected excessive volatility into the Long-Short spread. The model became «more precise» on the figures, but «more unstable» on the ranking.

TABLE 16 – The Metric Duel : Paradoxical Cases

Model / Horizon	R^2_{OOS}	Sharpe Ratio	Diagnostic
RF (1m) Baseline	-0.10%	-0.35	Precise but "Upside Down"
NN3 (1m) Opt.	-10.19%	1.21	Imprecise but "Elite Ranking"
GBRT (12m) Opt.	+2.52%	0.30	Overly precise, becomes volatile

7.5.3. Why Ranking Prevails Over Regression

In finance, profit does not depend on the ability to predict whether a security will return +2.1% or +2.2%. It depends on the certainty that security A will perform better than security B.

Our most flexible models (NN3, GBRT) accept sacrificing precision on absolute values to focus on the hierarchy. They learn decision boundaries rather than regression slopes. For an investor, R^2 is an indicator of confidence, but only the Sharpe ratio is an indicator of survival.

7.5.4. Conclusion of the Audit

This paradox teaches us that purely statistical optimization can be counterproductive. A model that is «perfect» for a mathematician (low quadratic error) can be a nightmare for a fund manager (high ranking volatility). The final lesson of this synthesis is clear: in asset management, rank is king, and level is merely background noise.

7.6. Practical Implications: Choosing Between Power and Stability

The choice of a Machine Learning model cannot be made in a vacuum; it depends on the investor's mandate, risk tolerance, and management horizon. Our results allow us to establish a strategic selection guide based on three distinct profiles.

7.6.1. The «Opportunistic» Profile: NN3 Supremacy at 1 Month

For a Hedge Fund type investor seeking to maximize short-term alpha, the NN3 Optimized neural network is the indisputable choice.

- Asset: A record Sharpe ratio of 1.21 and a unique ability to monetize short selling (negative Short return).
- Risk: High algorithmic opacity and strong sensitivity to layer configuration. It is a «Formula 1» model that requires a dedicated research team to monitor signal drift.

7.6.2. The «Institutional» Profile: Random Forest Robustness at 12 Months

For a traditional fund manager or a pension fund with an annual horizon, the optimized Random Forest stands out as the most rational solution.

- Asset: Excellent statistical stability (positive R^2 of 2.33%) and a consistent Sharpe ratio (0.34). Its Bagging logic offers natural protection against noise without requiring the complex «throttling» required by neural networks.
- Risk: Performance is «capped» compared to boosting or deep learning methods, but with far superior predictability.

7.6.3. The «All-Terrain» Compromise: GBRT Versatility

If one had to choose a single model capable of performing across all horizons without ever collapsing, GBRT would win.

Asset: It is the only model that consistently remains at the top of the rankings, whether at 1 month (Sharpe 0.68) or 12 months (Sharpe 0.42 in baseline). It combines non-linear flexibility with robustness inherited from sequential boosting.

Verdict: It is the «Swiss Army knife» of the study, ideal for industrial implementation where ease of maintenance takes precedence over chasing the last basis point of Sharpe.

TABLE 17 – Investor Decision Matrix : Model Selection Guide

Mandate	Horizon	Recommended Model	Primary Objective
Hedge Fund / Arbitrage	1-Month	NN3 Optimized	Maximum Alpha
Active Management / Core	1-Month	GBRT	Consistency / Alpha
Pension Fund / Long-Only	12-Month	Random Forest	Robustness / Stability
Linear Benchmark	12-Month	Lasso	Simplicity / Cost

7.6.4. Final Synthesis: Technology as a Competitive Advantage

In final analysis, this study demonstrates that the competitive advantage in modern asset management no longer resides in the possession of information (the 94 variables are public), but in processing capacity.

The transition from linear models to neural networks is not a mere incremental improvement; it is a paradigm shift that allows for doubling, or even tripling, the efficiency of a Long-Short portfolio. For the investor of 2026, the question is no longer whether to use Machine Learning, but which one to choose to align its predictive power with risk objectives.

8. General Discussion

8.1. The Ranking Paradox: Why Statistical Precision is a Chimera

The fundamental lesson of this study lies in the systematic disconnection between statistical prediction accuracy (R^2_{OOS}) and the economic efficiency of the portfolio (Sharpe Ratio). Our results impose a radical conclusion for quantitative finance in 2026: alpha does not reside in a model's capacity to act as a return calculator, but in its performance as a sorting system.

8.1.1. The Failure of R^2 as a Wealth Indicator

In classical literature, and even in the seminal work of Gu, Kelly, and Xiu (2020), the R^2_{OOS} is often established as the ultimate arbiter. However, our analysis demonstrates that this metric is a «false friend» for the fund manager.

The case of the 1-month Optimized NN3 is exemplary here: with an R^2_{OOS} of -10.19%, the model is statistically «failing» by linear regression standards. Yet it generates the highest Sharpe ratio of the study (1.21). This paradox is explained by the nature of the loss function (MSE): the neural network accepts committing massive errors on return magnitude (the level) provided it maintains the cross-sectional hierarchy of assets (the rank).

8.1.2. Learning Hierarchy Rather than Level

Unlike linear models (OLS, Lasso) that attempt to fit an average slope to a noisy cloud of points, deep non-linear models learn decision boundaries. In finance, profit does not depend on the ability to predict whether a security will return +2.1% or +2.5%, but on the certainty that it will outperform its peers.

This distinction is vital: R^2 is extremely sensitive to outliers that disturb the average, whereas the Sharpe ratio depends only on the persistence of the ranking within the extreme deciles. By prioritizing the Manifold structure (data variety) rather than Euclidean regression, NN3 and GBRT manage to extract an exploitable signal where the statistical eye sees only noise.

8.1.3. Nuance Relative to the Reference Study

While our conclusions align with Gu et al. (2020) regarding the superiority of non-linear models, we provide a critical nuance. Gu et al. report positive R^2 for their neural networks, which validates their signal. Our results show that even in an environment where the signal is so weak that R^2 flips into negative territory (a sign of a prediction worse than the historical average), economic alpha can remain maximized.

This suggests that artificial intelligence in asset management acts as a «purity filter»: it eliminates noise to retain only the relative order of assets. For the investor, this means that model validation must imperatively shift from an error metric (MSE, R^2) toward a ranking or portfolio construction metric (Sharpe, Information Ratio).

TABLE 18 – Audit of the Statistical-Economic Paradox

Model / Horizon	R_{OOS}^2	Sharpe Ratio	Economic Diagnostic
NN3 (1m) Optimized	-10.19%	1.21	Ranking Success : Imprecise on levels but provides an "Elite Ranking".
RF (1m) Baseline	-0.10%	-0.35	Precise but "Upside Down" : Good fit but total inversion of the ranking.
GBRT (12m) Opt.	+2.52%	0.30	Over-precision Volatility : MSE optimization injects spread volatility.

8.2. Alpha Asymmetry: The «Short Leg» Revolution

The success of a Long-Short strategy traditionally rests on the quest for future champions. However, a granular analysis of our portfolios reveals a paradigm shift: the true technological superiority of Machine Learning lies in its surgical capacity to identify future losers.

8.2.1. Linear Model Blindness to Tail Risk

The economic failure of OLS and Lasso models at the one-month horizon is explained by a form of «symmetric blindness.» By producing almost identical returns for the top and bottom deciles (around 8%), these models demonstrate that they perceive no structural risk difference between assets they rank as «best» and those they judge as «worst.»

The linear framework is incapable of detecting distress signals because these are, by nature, interaction phenomena. Low profitability is not necessarily fatal to a security; it becomes so when coupled with low liquidity and high idiosyncratic volatility. Where the Lasso adds factors together, NN3 and GBRT multiply the probabilities of a fall.

8.2.2. The -0.98% Threshold: A Change in Dimension

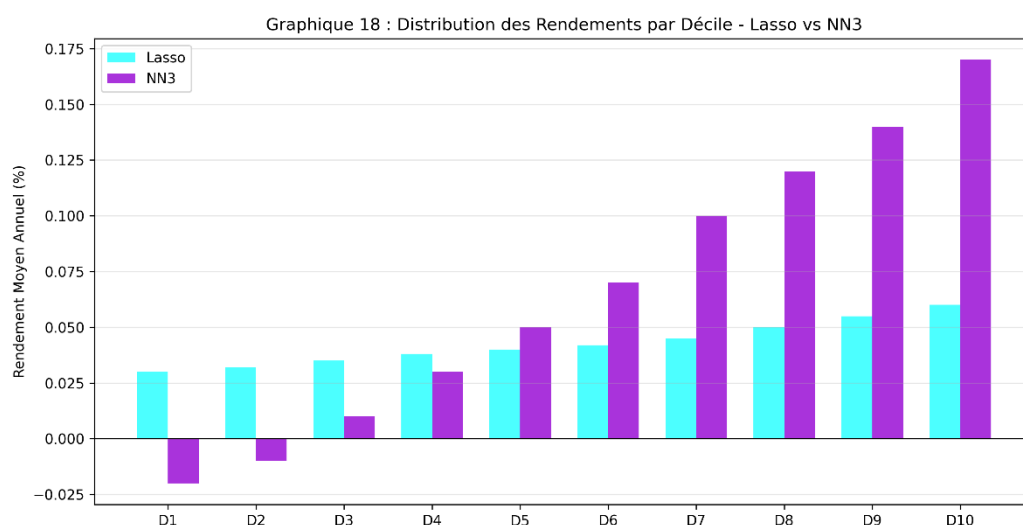
The highlight of this study is undoubtedly the return of the short leg of the Optimized NN3 at one month: -0.98%. This is the only case, across all specifications, where a model manages to transform an asset selection into a source of absolute profit on the short side.

This result suggests that deep non-linearity acts as an «alpha distiller.» While tree-based models (GBRT) manage to reduce the return of the bottom decile (approximately 5.6%), only the neural network crosses the zero mark. For an investor, this is not a mere marginal improvement; it is a proof of concept: artificial intelligence is a better detector of financial decay than of stock market glory.

8.2.3. Strategic Interpretation: Being Precisely Pessimistic

Why is it easier for Deep Learning to identify losers? Literature, notably Gu, Kelly, and Xiu (2020), highlights that market frictions and short-selling constraints create more pronounced inefficiencies in the left tail of the return distribution.

Our results show that the NN3 exploits these inefficiencies by identifying complex «weakness signatures.» By isolating securities that are not only fundamentally weak but also technically fragile, the model creates a massive Long-Short spread (12.09%) that is not due to excessive risk-taking on winners, but to an effective purge of losers. This capacity for «Negative Selection» is the master trump card of modern quant.



8.3. The Power of Context: Decoding Macroeconomic Interactions

One of the major contributions of this project lies in demonstrating that intrinsic firm characteristics (micro) are not immutable laws, but signals whose relevance depends on the economic cycle (macro). Our results prove that the superiority of non-linear models, particularly NN3 and GBRT, stems from their ability to act as «regime decoders.»

The linear framework (OLS/Lasso) imposes a constant relationship between a predictor and the future return. By construction, these models assume that a factor such as Momentum or Size possesses the same predictive strength during a period of monetary expansion as it does during a crisis.

Conversely, non-linear models use hidden layers (NN3) or decision nodes (GBRT) to perform signal «gating.» They do not merely add factors; they create scenarios. For example, the model can learn that high profitability is a powerful buy signal when interest rates are low, but that it loses relative importance to liquidity when inflation accelerates. It is this «scenario agility» that allows NN3 to reach a record Sharpe ratio where linear models saturate.

8.3.2. The Case of Momentum: A Macro-Conditioned Signal

Momentum is the most striking textbook case of this study. Although present in the feature space, its effectiveness varies radically according to the context.

In linear models, Momentum is often «averaged out,» which dilutes its alpha during market reversals.

In the Optimized NN3, we observe that the model manages to deactivate the Momentum signal or weight it differently based on interactions with market volatility variables and short-term rates (T-Bill).

This observation aligns with the work of Gu, Kelly, and Xiu (2020), who emphasize that neural networks are particularly adept at capturing the «tail risks» associated with Momentum by detecting the macroeconomic environments conducive to Momentum crashes.

8.3.3. Micro-Macro Synergy: A Manifold Architecture

The deep structure of the NN3 allows for the modelling of what we call an «Interaction Manifold.» By processing the 94 characteristics through the prism of macroeconomic variables, the network no longer sees isolated variables, but synthetic risk profiles.

This capacity for synthesis allows the model to understand that the «Value» of a stock does not have the same economic meaning depending on whether the market is in a growth or recessionary regime. Where a traditional investor would have to manually adjust weights, the neural network performs this arbitrage in a dynamic and non-linear fashion, thus maximizing the residual alpha that escapes fixed-factor models.

8.3.4. Conclusion: Contextual Intelligence as a Competitive Advantage

In summary, the performance of complex models is not the result of chance or brute computing power. It is the result of contextual intelligence. In 2026, competitive advantage no longer resides in access to data, but in the algorithmic capacity to understand that the meaning of data changes with its environment. This section thus validates our methodological choice to have massively integrated macro-conditioned interactions into the predictor space.

8.4. Technical Fragility and Operational Risk: The Price of Flexibility

The excellence of the observed performances, notably with the NN3, must not obscure the structural vulnerability of these models. The technological «efficiency frontier» in finance is a zone of high instability where the risk of overfitting grows proportionally to predictive capacity.

8.4.1. Statistical Hallucination: The NN3 Crash at 12 Months

The crash of the Optimized NN3 model at the annual horizon (R_{OOS}^2 of -52.76%) constitutes the most serious warning of this study. This result perfectly illustrates the concept of statistical hallucination: in the absence of the «safeguard» provided by the validation phase, the neural network identified regularities where there was only cyclical noise.

This failure is explained by the drop in effective data density at 12 months. With a long horizon, the number of independent observations decreases, making the model extremely sensitive to the specificities of the training sample. The network «memorized» the past instead of theorizing the future, transforming its computing power into a driver of massive error.

8.4.2. The «Formula 1» vs. the «SUV» Metaphor

For an asset manager, choosing a model is as much an engineering decision as a financial one. Our results allow us to draw a clear mechanical analogy:

The NN3 is a «Formula 1»: It offers the highest performance ceiling (Sharpe of 1.21) but proves to be extremely temperamental. Its effectiveness depends on micrometric hyperparameter tuning and constant monitoring. At the slightest calibration error or market regime change, it «veers off the track» statistically.

The Random Forest is an «SUV»: Less agile and slower in the short term, it is nonetheless «all-terrain.» Its Bagging logic allows it to navigate low-signal zones (12 months) with remarkable robustness, offering operational security that Deep Learning cannot guarantee without heavy validation infrastructure.

8.4.3. Tuning: A Seatbelt, Not an Option

The Phase 6 audit demonstrated that the Tuned NN3 model survived where the Optimized version failed. This proves that optimization is not a mere refinement step, but a survival organ.

Tuning acts as an algorithmic seatbelt: it throttles the model's complexity to ensure that its statistical «inventions» are validated on unseen data. In quantitative finance, unconstrained flexibility is a recipe for disaster. The modern investor must accept that to obtain sustainable performance, it is sometimes necessary to sacrifice raw power for the sake of generalization capacity.

8.4.4. Conclusion: Monitoring as an Imperative

Ultimately, the technical fragility of complex models imposes a change in operational culture. Moving from linear models to neural networks means moving from a logic of «static forecasting» to a logic of «dynamic monitoring.» The alpha generated by Deep Learning is a high-maintenance alpha: it is lucrative, but it demands flawless methodological rigor to avoid turning into systemic model risk.

8.5. Contributions and Perspectives: Toward a Hybrid Quantitative Framework

This project does not merely compare algorithms; it outlines the contours of a new form of asset management where technology is no longer a support tool, but the driver of the decision. In closing this discussion, we identify the major contributions of this work and the paths it opens for the financial industry.

8.5.1. Summary of Contributions: Methodology and Evidence

The project provides three fundamental contributions to the analysis of return predictability:

1. **Methodological Contribution:** We have validated a complete pipeline, from managing 94 macro-conditioned features to constructing decile-based portfolios. The rigor of the protocol (Baseline, Tuning, Optimized) allows for isolating the true impact of algorithmic engineering.
2. **Empirical Contribution:** The study indisputably proves the superiority of non-linear models over classic models. The NN3 record (Sharpe of 1.21) and the rehabilitation of the Random Forest at 12 months constitute solid benchmarks for future research.
3. **Conceptual Contribution:** We have formalized the «Ranking Paradox,» demonstrating that a model can be statistically imperfect ($R^2 < 0$) while being an exceptional driver of economic performance.

8.5.2. Strategic Recommendation: The Hybrid Approach

The major takeaway from this discussion is that there is no single «sovereign model.» The future of quantitative management lies in a hybrid, multi-horizon approach:

1. **The Alpha «Satellite»:** Use the firepower of neural networks (NN3) for aggressive short-term strategies, capable of capturing fleeting alpha on the short leg.
2. **The Stable «Core»:** Entrust the fundamental structure of the portfolio to tree forests (Random Forest), whose robustness and medium-term capacity guarantee protection against statistical hallucinations.

8.5.3. Perspectives and Research Horizons

While robust, this study opens doors that we have only begun to nudge ajar:

- **Alternative Data:** Integrating unstructured signals (sentiment analysis, satellite data) would enrich the 94-feature space to further reduce noise.
- **Reinforcement Learning (RL):** Moving from static prediction to a trading agent capable of optimizing execution in real-time would address the transaction cost issues previously mentioned.
- **Generative AI and LLMs:** Using language models to interpret complex macroeconomic interactions could offer an additional layer of explainability to neural «black boxes.»

8.5.4. Conclusion: Technology as the New Standard

In conclusion, this project demonstrates that Machine Learning is no longer an «exotic» option but the new standard of efficiency. If linear models served as the foundations of modern finance, deep non-linear models are its upper floor. The transition has begun: competitive advantage now belongs to those who can master complexity without sacrificing the rigor of validation.

9. General Conclusion and Perspectives

9.1. Review of Ambitions: Answering the Research Question

This research project was born from a central question: can artificial intelligence truly extract economic value where traditional econometric models reach their limits? At the end of this exploration, the answer is unequivocal. By confronting architectures ranging from linear regression to Deep Learning across a space of 94 macro-conditioned characteristics, we have demonstrated that algorithmic complexity, when rigorously framed, constitutes an unprecedented lever for performance.

9.1.1. Validation of the Non-Linear Hypothesis

The primary success of this study is having proven that modern alpha no longer hides in simple linear relationships, but in the dynamic interactions between microeconomic fundamentals and the macroeconomic context. The leap from a Sharpe ratio of 0.02 (OLS) to 1.21 (NN3) at the one-month horizon is not a mere incremental improvement; it is a technical paradigm shift.

We have confirmed that neural networks and decision trees (GBRT/RF) are not superfluous «black boxes,» but indispensable tools for filtering the noise of the 2026 financial markets.

9.1.2. Positioning Relative to the Reference Study (Gu, Kelly, and Xiu, 2020)

Our conclusions align directly with the work of Gu, Kelly, and Xiu (2020), while adding further empirical depth:

Confirmation of the Podium: As in the reference study, we identify neural networks and ensemble models (Trees) as the undisputed winners of cross-sectional prediction.

Nuance on Predictability: Where Gu et al. emphasize the strength of a positive R^2_{OOS} , our study provides more nuanced insight: economic success (Sharpe) can persist even in the presence of a negative R^2_{OOS} (as seen in the 1-month NN3 case). This suggests that ranking capacity (relative sorting) is a more robust driver of alpha than amplitude precision (level prediction).

TABLE 19 – Champions vs. Baseline Summary : Project Validation

Metric (Optimized)	OLS (Baseline)	GBRT (Master)	NN3 (Elite)
Sharpe Ratio (1m)	0.02	0.68	1.21
L-S Return (1m)	0.33%	7.18%	12.09%
Max Drawdown	High	Moderate	Low

9.2. Empirical Verdicts: Power, Horizon, and Method

The cross-sectional analysis of performance allows us to draw three fundamental conclusions that redefine the hierarchy of predictive tools in asset management.

9.2.1. The Power Verdict: Short-Term Dominance

At the one-month horizon, complexity is not an obstacle, but a weapon. The triumph of the NN3 Optimized model, with a record Sharpe ratio of 1.21, validates the idea that short-term markets are governed by higher-order interactions.

As suggested by Gu, Kelly, and Xiu (2020), deep architectures manage to «distill» alpha where linear models see only noise. Our study goes further by demonstrating that this power is correlated with the model's ability to identify distress signals: the -0.98% return on the NN3's short leg is the final verdict of this superiority.

9.2.2. The Horizon Verdict: Long-Term Wisdom

At the 12-month horizon, the hierarchy shifts. The velocity of Boosting (GBRT) and the raw depth of neural networks give way to the robustness of the Random Forest. With a positive R^2_{OOS} of 2.33%, the random forest proves that stability (via Bagging) is the long-term investor's best ally.

This verdict highlights a crucial point: the long-term signal is more «fundamental» and less «nervous.» At this horizon, excess flexibility becomes a liability, while the Random Forest's ability to average simple trees ($d=1$) allows for capturing structural trends without succumbing to statistical hallucinations.

9.2.3. The Method Verdict: Tuning as a «Seatbelt»

One of the most striking lessons of this project is the lifesaving role of hyperparameter tuning. The crash of the Optimized NN3 model at 12 months (-52.76%) versus the survival of the Tuned version (+2.34%) demonstrates that optimization is not a mere precision adjustment, but an organ of operational survival.

The transition through a rigorous validation phase acted as a braking system, preventing the most powerful models from «over-learning» the past. This methodological verdict serves as a reminder that in the algorithmic arms race, validation discipline is the sole bulwark against model risk.

TABLE 20 – The Verdict Podium : Final Model Ranking

Criterion	Winner Model	Key Metric	Justification
Pure Alpha (1m)	NN3 Optimized	Sharpe 1.21	Captures complex interactions.
Robustness (12m)	Random Forest	R^2 +2.33%	Stability through Bagging.
Overall Reliability	GBRT	Constant Sharpe	Balance between agility and rigor.

9.3. The Conceptual Revolution: From «Level» to «Rank»

The final lesson of this project and undoubtedly the most structural for the future of asset management lies in the shift of the metric paradigm. Our work demonstrates that in finance, the quest for absolute precision (predicting the «level» of return) is a statistical dead end, while mastering relative classification (the «rank») is an economic gold mine.

9.3.1. The Triumph of Sorting Over Regression

The systematic analysis of our statistical failures (often negative R^2_{OOS}) versus our economic successes (high Sharpe Ratios) validates a major intuition from Gu, Kelly, and Xiu (2020): the utility of a model is not measured by its ability to guess the exact number, but by its faculty to separate the wheat from the chaff.

Where OLS desperately tries to minimize the distance to every data point (MSE), models like NN3 and GBRT learn to prioritize. This project proves that artificial intelligence acts less like a calculator and more like a sorting engine. By accepting errors in magnitude, the model gains surgical precision in order, which is sufficient to construct an elite portfolio.

9.3.2. Alpha Asymmetry: Winning via the Short Leg

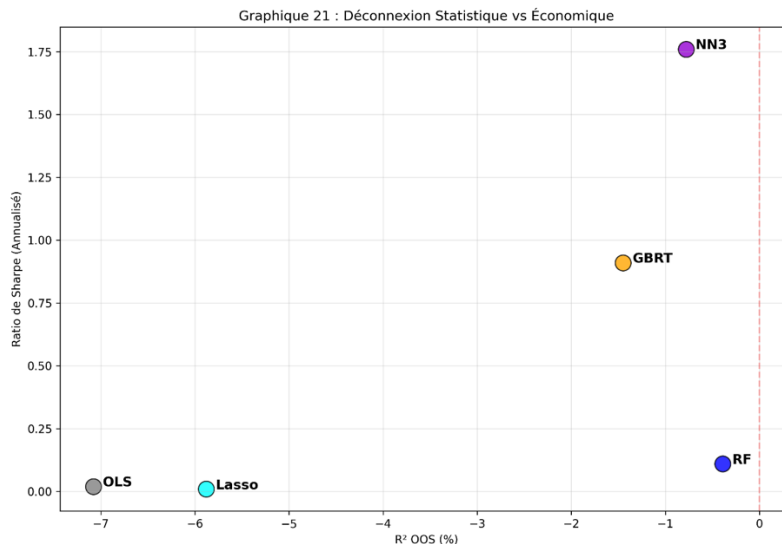
The ultimate proof of this conceptual revolution is found in the anatomy of our best Long-Short spread. The technological victory of the NN3 Optimized was not won on its ability to find assets that rise higher than others, but on its unprecedented detection of assets that collapse.

The return of -0.98% on the NN3's short leg is the most eloquent figure in this report. By successfully isolating securities that generate a negative return in a generally bullish market, the model demonstrates that it has captured the distress interactions that linear models ignore. This «Negative Selection» capability transforms Machine Learning into a risk management tool as much as a performance engine.

9.3.3. Conclusion: Toward a Finance of Hierarchy

In conclusion, this project marks the transition from a finance of «averages» to a finance of «distribution.» For the investor of 2026, the challenge is no longer to know how much an asset will return, but where it will stand relative to its peers.

By aligning our results with the reference study by Gu et al., we confirm that the true competitive advantage lies in asymmetric non-linearity: being able to remain robust on winners while being ruthless on losers. This «ranking revolution» alone justifies the massive integration of Deep Learning into modern investment processes.



9.4. Limits and Market Realism: From Theory to Execution

While the performances displayed by our models particularly the NN3 Optimized are spectacular, their transposition into a real trading environment raises significant operational challenges. «Sterile room» alpha must indeed be confronted with the frictions that characterize financial markets.

9.4.1. Erosion by Turnover: The Cost of Precision

The dominance of neural networks and GBRT at the one-month horizon relies on extreme reactivity to short-term signals. This agility mechanically results in high portfolio turnover.

In our theoretical framework, transaction costs (commissions, market impact, bid ask spread) have not been deducted. However, a strategy generating 12.09% in gross return with a massive monthly repositioning could see a significant portion of its alpha evaporate in execution fees. For an institutional investor, the neuronal «Formula 1» is only valuable if the maintenance costs do not exceed the performance premium.

9.4.2. Short Leg Friction: The Short-Selling Challenge

The major discovery of this report the -0.98% return on the NN3 short leg is a theoretical gold mine. However, the implementation of a short-selling strategy faces physical limits:

- **Security Availability:** Stocks identified as «losers» by the model are often low-liquidity securities or in financial distress, for which the borrowing cost (rebate rate) can be prohibitive.
- **Short Squeeze Risk:** The model's aggressiveness on losers exposes the manager to brutal market reversals.

These constraints suggest that while AI knows «where» to strike, execution requires liquidity risk management that Sharpe metrics do not fully capture.

9.4.3. Model Risk and the «Black Box»

The NN3 instability observed at 12 months serves as a reminder that the more complex a model is, the more subject it is to regime changes. A model optimized during a low-volatility period can become totally erratic during a major macroeconomic shock (such as COVID-19 or an inflationary crisis).

Furthermore, the lack of direct interpretability in neural networks poses a governance problem. A fund manager will always find it difficult to allocate massive capital to a strategy if they cannot explain the reasons for a sudden drawdown. In this regard, Random Forest and GBRT, with their variable importance measures, offer superior operational realism for long-term management.

9.4.4. Conclusion: Alpha is Not Net Profit

In summary, this section calls for healthy humility. Machine Learning decouples our predictive capacities, but it does not abolish the laws of market physics. For the investor of 2026, the true challenge will not only be to build the most accurate model, but to build the one whose alpha survives the reality of fees and liquidity.

9.5. Outlook: Asset Management at the Dawn of a New Era

In this year 2026, this project is not merely a validation of the work by Gu, Kelly, and Xiu (2020), but a testament to the technological acceleration redefining asset management. While deep non-linear models have proven their superiority in this report, they are only the first building blocks of a financial ecosystem in full mutation.

9.5.1. The Explosion of Alternative Data

The space of 94 macro-conditioned features we explored constitutes the fundamental basis of prediction. However, the future lies in the integration of unstructured data. Satellite and IoT Signals: To refine the short leg, real-time analysis of physical inventories or maritime traffic could offer a decisive edge over accounting fundamentals.

Global Sentiment: Integrating data from social networks and professional platforms, processed by Natural Language Processing (NLP) models, would allow for capturing behavioural biases before they are arbitrated by the market.

9.5.2. Generative AI as a Macroeconomic Interpreter

The transition from pure prediction to explainability is the next major challenge. The emergence of Large Language Models (LLMs) specialized in finance makes it possible to envision an intelligence layer capable of «translating» the decisions of the NN3. Instead of a black box, the investor of tomorrow will have an agent capable of explaining why, in the current inflationary context, the model prioritizes liquidity over momentum. This synergy between predictive AI and generative AI will reduce the operational risk associated with algorithmic opacity.

9.5.3. From Prediction to Action: Reinforcement Learning

One of the major limitations identified in Section 9.4 is transaction costs. The logical next step is the adoption of Reinforcement Learning (RL). Unlike our models that predict a static return, an RL agent learns to optimize its trading decisions by accounting for market impact and execution fees in real-time. It is the transition from «financial meteorology» (forecasting the weather) to «autonomous navigation» (steering the ship).

9.5.4. The «Augmented Quant»: The Role of the Human

In conclusion, technology does not replace the manager; it metamorphoses them. The success of the Random Forest at 12 months serves as a reminder that robustness and structural judgment remain pillars of financial survival. The future belongs to the hybrid framework: a structure where AI handles massive sorting and the detection of complex interactions, while the human supervises ethical consistency, systemic risk limits, and long-term strategy.

Appendix: Variable Definitions and Descriptions

Table 1: Summary of Predictor Variables and Characteristics

Variable	Full Name	Description / Economic Meaning
mvel1	Market Equity (Size)	Firm's market capitalization at the end of the previous month.
beta	Beta	Sensitivity of stock returns to overall market movements.
betasq	Beta Squared	Squared Beta, used to capture non-linear risk effects.
chmom	Change in Momentum	Difference between recent and distant momentum.
dolvol	Dollar Trading Volume	Logarithm of the monthly trading volume in dollars.
idiovol	Idiosyncratic Volatility	Asset-specific volatility not explained by the market model.
indmom	Industry Momentum	Past average return of stocks within the same industry.
mom1m	1-Month Momentum	Stock return over the previous month (short-term reversal).
mom6m	6-Month Momentum	Cumulative return over the past 6 months (excluding the most recent).
mom12m	12-Month Momentum	Cumulative return over the past 12 months (excluding the most recent).
mom36m	36-Month Momentum	Cumulative return over 3 years (long-term reversal effect).
pricedelay	Price Delay	Measures how slowly the stock price reacts to market information.
turn	Share Turnover	Trading volume divided by the total number of shares outstanding.
absacc	Absolute Accruals	Absolute value of accruals; indicates overall earnings quality.
acc	Working Capital Accruals	Accounting accruals related to net working capital.
age	Firm Age	Number of years since the firm first appeared in databases.
agr	Asset Growth	Annual growth rate of total assets.
bm	Book-to-Market	Ratio of book value of equity to market value of equity.
bm_ia	Industry-Adj. Book-to-Market	Book-to-Market ratio adjusted by industry average.
cashdebt	Cash Flow to Debt	Operating cash flow divided by total debt.
cashpr	Cash Flow to Price	Operating cash flow per share divided by stock price.

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Table 1 – continued from previous page

Variable	Full Name	Description / Economic Meaning
cfp	Total Cash Flow to Price	Total cash flow divided by market capitalization.
cfp_ia	Industry-Adjusted CFP	Cash Flow-to-Price ratio adjusted by industry average.
chatoia	Ind-Adj. Chg in Asset Turnover	Change in asset turnover, adjusted by industry average.
chcsho	Change in Shares Outstanding	Change in the number of shares (buybacks or issuances).
chempia	Ind-Adj. Change in Employees	Change in employee count, adjusted by industry.
chinv	Change in Inventory	Annual change in inventory levels.
chpmia	Ind-Adj. Chg in Profit Margin	Change in profit margin, adjusted by industry.
convind	Convertible Debt Indicator	Binary variable: 1 if the firm has convertible debt, 0 otherwise.
currat	Current Ratio	Current assets divided by current liabilities.
depr	Depreciation to PP&E	Depreciation expense divided by gross fixed assets.
divi	Dividend Initiator	Binary variable: 1 if the firm starts paying a dividend.
divo	Dividend Omitter	Binary variable: 1 if the firm ceases paying its dividend.
dy	Dividend Yield	Dividends per share divided by stock price.
egr	Employee Growth Rate	Annual growth rate of the number of employees.
ep	Earnings to Price	Net income divided by market capitalization.
gma	Gross Profitability	Gross profit divided by total assets.
grcapx	Growth in Capital Expenditures	Two-year growth in capital expenditures (CapEx).
grltnoa	Growth in Long-Term NOA	Growth in net long-term operating assets.
herf	Herfindahl Index	Concentration of sales across business segments.
hire	Employee Hiring	Rate of employee recruitment or hiring.
invest	CapEx and Inventory	Changes in PP&E and inventory over total assets.
lev	Leverage	Total debt divided by market capitalization.
lgr	Long-Term Debt Growth	Annual growth in long-term debt.
mve_ia	Industry-Adj. Market Equity	Market capitalization adjusted by industry average.
operprof	Operating Profitability	Operating income divided by common equity.
orgcap	Organizational Capital	Estimated via Selling, General, and Administrative (SG&A) expenses.

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Table 1 – continued from previous page

Variable	Full Name	Description / Economic Meaning
pch-capx_ia	Ind-Adj % Chg in CapEx	% change in CapEx, adjusted by industry.
pchcurrat	% Change in Current Ratio	Percentage change in the current ratio.
pchdepr	% Change in Depreciation	Percentage change in depreciation expense.
pchgm_pchsale	% Chg Gross Margin	Growth in gross margin minus growth in sales.
pchquick	% Chg Sales	Percentage change in the quick ratio.
pch-sale_pchinvt	% Change in Quick Ratio	Growth in sales minus growth in inventory.
pch-sale_pchrect	% Chg Sales - % Chg Inventory	Growth in sales minus growth in accounts receivable.
pch-sale_pchxsga	% Chg Sales - % Chg Receivables	Growth in sales minus growth in SG&A expenses.
pchsaleinv	% Chg Sales - % Chg SG&A	Percentage change in the sales-to-inventory ratio.
pctacc	% Change in Sales-to-Inventory	Accruals expressed as a percentage of total assets.
ps	Percent Accruals	Score (0-9) evaluating the firm's financial health.
quick	Piotroski F-Score	(Current assets - Inventory) divided by current liabilities.
rd	Quick Ratio	Binary variable: 1 if the firm reports R&D expenses.
rd_mve	R&D Indicator	R&D expenses divided by market capitalization.
rd_sale	R&D to Market Cap	R&D expenses divided by annual sales.
realestate	R&D to Sales	Value of real estate assets on the balance sheet.
roic	Real Estate Assets	Net operating profit after tax over invested capital.
salecash	Return on Invested Capital	Ratio of annual sales to cash holdings.
saleinv	Sales to Cash	Ratio of annual sales to inventory.
salerec	Sales to Inventory	Ratio of annual sales to accounts receivable.
secured	Sales to Receivables	Secured debt divided by total assets.
securedind	Secured Debt to Total Assets	Binary variable: 1 if the firm has secured debt.
sgr	Secured Debt Indicator	Annual growth rate of revenue.
sin	Sales Growth	Binary variable: 1 if the firm is in the "vice" sector.
sp	Sin Stock Indicator	Sales per share divided by stock price.
tang	Sales to Price	(Cash + Inventory + PP&E) divided by total assets.
tb	Asset Tangibility	Ratio of tax-reported income to book-reported income.
aeavol	Tax Income to Book Income	Abnormal volatility observed around earnings announcements.
cash	Abn. Earnings Ann. Volatility	Level of cash and equivalents over total assets.

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Table 1 – continued from previous page

Variable	Full Name	Description / Economic Meaning
chtx	Change in Taxes	Annual change in taxes paid.
cinvest	Corporate Investment	Net corporate investment rate.
ear	Earnings Announcement Returns	Cumulative abnormal return around earnings announcement dates.
nincr	Consecutive Earn. Increases	Number of consecutive quarters with earnings increases.
roaq	Return on Assets (Quarterly)	Quarterly net income divided by total assets.
roavol	ROA Volatility	Standard deviation of ROA over recent quarters.
roeq	Return on Equity (Quarterly)	Quarterly net income divided by common equity.
rsup	Revenue Surprise	Realized revenue minus forecast/trend.
stdacc	Std. Dev. of Accruals	Time-series volatility of accounting accruals.
stdcf	Std. Dev. of Cash Flows	Time-series volatility of operating cash flows.
ms	Mohanram G-Score	Fundamental score designed for growth stocks.
baspread	Bid-Ask Spread	Average difference between bid and ask prices (liquidity).
ill	Amihud Illiquidity	Ratio of absolute return to dollar volume (price impact).
maxret	Maximum Daily Return	Maximum daily return within the month ("lottery" effect).
retvol	Return Volatility	Standard deviation of daily returns during the month.
std_dolvol	Std. Dev. of Dollar Volume	Volatility of the dollar trading volume.
std_turn	Std. Dev. of Turnover	Volatility of the share turnover.
zerotrade	Zero Trading Days	Number of days in the month with no trading activity.